

EXAMPLE 8 Find the kernel of the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ where $T([x_1, x_2, x_3]) = [x_1 - 2x_2, x_1 + 4x_3]$.

SOLUTION We simply find the nullspace of the standard matrix representation A of T . Writing down and then reducing the matrix A , we have

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 1 & 0 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 \\ 0 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \end{bmatrix}.$$

Thus we find that $\ker(T) = \text{sp}([-4, -2, 1])$. ■

Matrix Operations and Linear Transformations

It is very fruitful to be able to hop back and forth at will between matrices and their associated linear transformations. Every property of matrices has an interpretation for linear transformations, and vice versa. For example, the rank equation, $\text{rank}(A) + \text{nullity}(A) = n$, for an $m \times n$ matrix A becomes

$$\dim(\text{range } T) + \dim(\ker(T)) = \dim(\text{domain } T).$$

The dimension of $\text{range } T$ is called the **rank** of T , and the dimension of $\ker(T)$ is called the **nullity** of T .

Also, matrix multiplication and matrix inversion have very significant analogues in terms of transformations. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T': \mathbb{R}^m \rightarrow \mathbb{R}^k$ be two linear transformations. We can consider the **composite** function $(T' \circ T): \mathbb{R}^n \rightarrow \mathbb{R}^k$ where $(T' \circ T)(\mathbf{x}) = T'(T(\mathbf{x}))$ for $\mathbf{x} \in \mathbb{R}^n$. Figure 2.4 gives a graphic illustration of this composite map.

Now suppose that A is the $m \times n$ matrix associated with T and that B is the $k \times m$ matrix associated with T' . Then we can compute $T'(T(\mathbf{x}))$ as

$$T'(T(\mathbf{x})) = T'(A\mathbf{x}) = B(A\mathbf{x}).$$

But

$$B(A\mathbf{x}) = (BA)\mathbf{x}, \quad \text{Associativity of matrix multiplication}$$

so $(T' \circ T)(\mathbf{x}) = (BA)\mathbf{x}$. From Example 3, we see that $T' \circ T$ is again a linear transformation, and that the matrix associated with it is the product of the matrix associated with T' and the matrix associated with T , in that order. Notice how easily this follows from the associativity of matrix multiplication.

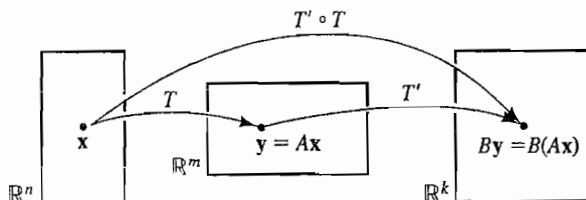


FIGURE 2.4
The composite map $T' \circ T$.

It really makes us appreciate the power of associativity! We can also show directly from Definition 2.3 that the composite of two linear transformations is again a linear transformation. (See Exercise 31.)

Matrix Multiplication and Composite Transformations

A composition of two linear transformations T and T' yields a linear transformation $T' \circ T$ having as its associated matrix the product of the matrices associated with T' and T , in that order.

This result has some surprising uses.

ILLUSTRATION 1 (*The Double Angle Formulas*) It is shown in the next section that rotation of the plane $\mathbb{R}^2$ counterclockwise about the origin through an angle θ is a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with standard matrix representation

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad \text{Counterclockwise rotation through } \theta \quad (3)$$

Thus, T applied twice—that is, $T \circ T$ —rotates the plane through 2θ . Replacing θ by 2θ in matrix (3), we find that the standard matrix representation for $T \circ T$ must be

$$\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}. \quad (4)$$

On the other hand, we know that the standard matrix representation for the composition $T \circ T$ must be the square of the standard matrix representation for T , and so matrix (4) must be equal to

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta - \sin^2 \theta & -2 \sin \theta \cos \theta \\ 2 \sin \theta \cos \theta & -\sin^2 \theta + \cos^2 \theta \end{bmatrix}. \quad (5)$$

Comparing the entries in matrix (4) with the final result in Eq. (5), we obtain the double angle trigonometric identities

$$\sin 2\theta = 2 \sin \theta \cos \theta \quad \text{and} \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta. \quad \blacksquare$$

Let us see how matrix invertibility reflects a corresponding property of the associated linear transformation. Suppose that A is an invertible $n \times n$ matrix, and let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the associated linear transformation, so that $y = T(x) = Ax$. There exists a linear transformation of $\mathbb{R}^n$ into $\mathbb{R}^n$ associated with A^{-1} ; we denote this by T^{-1} , so that $T^{-1}(y) = A^{-1}y$. The matrix of the composite transformation $T^{-1} \circ T$ is the product $A^{-1}A$, as indicated in the preceding box. Because $A^{-1}A = I$ and $Ix = x$, we see that $(T^{-1} \circ T)(x) = x$. That is, $T^{-1} \circ T$ is the identity transformation, leaving all vectors fixed. (See Fig. 2.5.) Because $AA^{-1} = I$ too, we see that $T \circ T^{-1}$ is also the identity transformation on $\mathbb{R}^n$. If

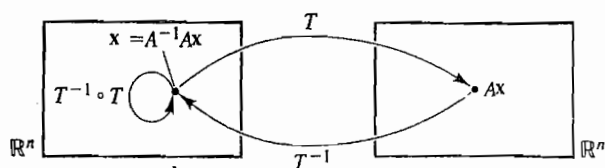


FIGURE 2.5

$T^{-1} \circ T$ is the identity transformation.

$y = T(x)$, then $x = T^{-1}(y)$. This transformation T^{-1} is the **inverse transformation** of T , and T is an **invertible** linear transformation.

Invertible Matrices and Inverse Transformations

Let A be an invertible $n \times n$ matrix with associated linear transformation T . The transformation T^{-1} associated with A^{-1} is the inverse transformation of T , and $T \circ T^{-1}$ and $T^{-1} \circ T$ are both the identity transformation on $\mathbb{R}^n$. A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible if and only if its associated matrix is invertible.

EXAMPLE 9 Show that the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T([x_1, x_2, x_3]) = [x_1 - 2x_2 + x_3, x_2 - x_3, 2x_2 - 3x_3]$ is invertible, and find a formula for its inverse.

SOLUTION Using column-vector notation, we see that $T(x) = Ax$, where

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & -3 \end{bmatrix}.$$

Next, we find the inverse of A :

$$\left[\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & -3 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 2 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & -2 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 4 & -1 \\ 0 & 1 & 0 & 0 & 3 & -1 \\ 0 & 0 & 1 & 0 & 2 & -1 \end{array} \right].$$

Therefore,

$$T^{-1}(x) = A^{-1}x = \begin{bmatrix} 1 & 4 & -1 \\ 0 & 3 & -1 \\ 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + 4x_2 - x_3 \\ 3x_2 - x_3 \\ 2x_2 - x_3 \end{bmatrix},$$

which we express in row notation as

$$T^{-1}([x_1, x_2, x_3]) = [x_1 + 4x_2 - x_3, 3x_2 - x_3, 2x_2 - x_3].$$

In Exercise 30, we ask you to verify that $T^{-1}(T(x)) = x$, as in Figure 2.5. ■

SUMMARY

1. A function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a *linear transformation* if $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(r\mathbf{u}) = rT(\mathbf{u})$ for all vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and all scalars r .
2. If A is an $m \times n$ matrix, then the function $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $T_A(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$ is a linear transformation.
3. A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is uniquely determined by $T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_n)$ for any basis $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ of $\mathbb{R}^n$.
4. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and let A be the $m \times n$ matrix whose j th column vector is $T(\mathbf{e}_j)$. Then $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$; the matrix A is the standard matrix representation of T . The kernel of T is the nullspace of A , and the range of T is the column space of A .
5. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T': \mathbb{R}^m \rightarrow \mathbb{R}^k$ be linear transformations with standard matrix representations A and B , respectively. The composition $T' \circ T$ of the two transformations is a linear transformation, and its standard matrix representation is BA .
6. If $\mathbf{y} = T(\mathbf{x}) = A\mathbf{x}$ where A is an invertible $n \times n$ matrix, then T is invertible and the transformation T^{-1} defined by $T^{-1}(\mathbf{y}) = A^{-1}\mathbf{y}$ is the inverse of T . Both $T^{-1} \circ T$ and $T \circ T^{-1}$ are the identity transformation of $\mathbb{R}^n$.

EXERCISES

1. Is $T([x_1, x_2, x_3]) = [x_1 + x_2, x_1 - 3x_2]$ a linear transformation of $\mathbb{R}^3$ into $\mathbb{R}^2$? Why or why not?
2. Is $T([x_1, x_2, x_3]) = [0, 0, 0, 0]$ a linear transformation of $\mathbb{R}^3$ into $\mathbb{R}^4$? Why or why not?
3. Is $T([x_1, x_2, x_3]) = [1, 1, 1, 1]$ a linear transformation of $\mathbb{R}^3$ into $\mathbb{R}^4$? Why or why not?
4. Is $T([x_1, x_2]) = [x_1 - x_2, x_2 + 1, 3x_1 - 2x_2]$ a linear transformation of $\mathbb{R}^2$ into $\mathbb{R}^3$? Why or why not?
5. If $T([1, 0]) = [3, -1]$ and $T([0, 1]) = [-2, 5]$, find $T([4, -6])$.
6. If $T([-1, 0]) = [2, 3]$ and $T([0, 1]) = [5, 1]$, find $T([-3, -5])$.
7. If $T([1, 0, 0]) = [3, 1, 2]$, $T([0, 1, 0]) = [2, -1, 4]$, and $T([0, 0, 1]) = [6, 0, 1]$, find $T([2, -5, 1])$.
8. If $T([1, 0, 0]) = [-3, 1]$, $T([0, 1, 0]) = [4, -1]$, and $T([0, -1, 1]) = [3, -5]$, find $T([-1, 4, 2])$.
9. If $T([-1, 2]) = [1, 0, 0]$ and $T([2, 1]) = [0, 1, 2]$, find $T([0, 10])$.
10. If $T([-1, 1]) = [2, 1, 4]$ and $T([1, 1]) = [-6, 3, 2]$, find $T([x, y])$.
11. If $T([1, 2, -3]) = [1, 0, 4, 2]$, $T([3, 5, 2]) = [-8, 3, 0, 1]$, and $T([-2, -3, -4]) = [0, 2, -1, 0]$, find $T([5, -1, 4])$.
[Computational aid: See Example 4 in Section 1.5.]
12. If $T([2, 3, 0]) = 8$, $T([1, 2, -1]) = -5$, and $T([4, 5, 1]) = 17$, find $T([-3, 11, -4])$.
[Computational aid: See the answer to Exercise 7 in Section 1.5.]

In Exercises 5–12, assume that T is a linear transformation. Refer to Example 7 for Exercises 9–12, if necessary.

In Exercises 13–18, the given formula defines a linear transformation. Give its standard matrix representation.

13. $T([x_1, x_2]) = [x_1 + x_2, x_1 - 3x_2]$
14. $T([x_1, x_2]) = [2x_1 - x_2, x_1 + x_2, x_1 + 3x_2]$
15. $T([x_1, x_2, x_3]) = [x_1 + x_2 + x_3, x_1 + x_2, x_1]$
16. $T([x_1, x_2, x_3]) = [2x_1 + x_2 + x_3, x_1 + x_2 + 3x_3]$
17. $T([x_1, x_2, x_3]) = [x_1 - x_2 + 3x_3, x_1 + x_2 + x_3, x_1]$
18. $T([x_1, x_2, x_3]) = x_1 + x_2 + x_3$
19. If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is defined by $T([x_1, x_2]) = [2x_1 + x_2, x_1, x_1 - x_2]$ and $T': \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T'([x_1, x_2, x_3]) = [x_1 - x_2 + x_3, x_1 + x_2]$, find the standard matrix representation for the linear transformation $T' \circ T$ that carries $\mathbb{R}^2$ into $\mathbb{R}^2$. Find a formula for $(T' \circ T)([x_1, x_2])$.
20. Referring to Exercise 19, find the standard matrix representation for the linear transformation $T \circ T'$ that carries $\mathbb{R}^3$ into $\mathbb{R}^3$. Find a formula for $(T \circ T')([x_1, x_2, x_3])$.

In Exercises 21–28, determine whether the indicated linear transformation T is invertible. If it is, find a formula for $T^{-1}(\mathbf{x})$ in row notation. If it is not, explain why it is not.

21. The transformation in Exercise 13.
22. The transformation in Exercise 14.
23. The transformation in Exercise 15.
24. The transformation in Exercise 16.
25. The transformation in Exercise 17.
26. The transformation in Exercise 18.
27. The transformation in Exercise 19.
28. The transformation in Exercise 20.
29. Mark each of the following True or False.
 - a. Every linear transformation is a function.
 - b. Every function mapping $\mathbb{R}^n$ into $\mathbb{R}^m$ is a linear transformation.
 - c. Composition of linear transformations corresponds to multiplication of their standard matrix representations.
 - d. Function composition is associative.

- e. An invertible linear transformation mapping $\mathbb{R}^n$ into itself has a unique inverse.
- f. The same matrix may be the standard matrix representation for several different linear transformations.
- g. A linear transformation having an $m \times n$ matrix as standard matrix representation maps $\mathbb{R}^n$ into $\mathbb{R}^m$.
- h. If T and T' are different linear transformations mapping $\mathbb{R}^n$ into $\mathbb{R}^m$, then we may have $T(\mathbf{e}_i) = T'(\mathbf{e}_i)$ for some standard basis vector $\mathbf{e}_i$ of $\mathbb{R}^n$.
- i. If T and T' are different linear transformations mapping $\mathbb{R}^n$ into $\mathbb{R}^m$, then we may have $T(\mathbf{e}_i) = T'(\mathbf{e}_i)$ for all standard basis vectors $\mathbf{e}_i$ of $\mathbb{R}^n$.
- j. If $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ is a basis for $\mathbb{R}^n$ and T and T' are linear transformations mapping $\mathbb{R}^n$ into $\mathbb{R}^m$, then $T(\mathbf{x}) = T'(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$ if and only if $T(\mathbf{b}_i) = T'(\mathbf{b}_i)$ for $i = 1, 2, \dots, n$.
30. Verify that $T^{-1}(T(\mathbf{x})) = \mathbf{x}$ for the linear transformation T in Example 9 of the text.
31. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T': \mathbb{R}^m \rightarrow \mathbb{R}^k$ be linear transformations. Prove directly from Definition 2.3 that $(T' \circ T): \mathbb{R}^n \rightarrow \mathbb{R}^k$ is also a linear transformation.
32. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Prove from Definition 2.3 that $T(r\mathbf{u} + s\mathbf{v}) = rT(\mathbf{u}) + sT(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and all scalars r and s .

Exercise 33 shows that the reduced row-echelon form of a matrix is unique.

33. Let A be an $m \times n$ matrix with row-echelon form H , and let V be the row space of A (and thus of H). Let $W_k = \text{sp}(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_k)$ be the subspace of $\mathbb{R}^n$ generated by the first k rows of the $n \times n$ identity matrix. Consider $T_k: V \rightarrow W_k$ defined by

$$T_k([x_1, x_2, \dots, x_n]) = [x_1, x_2, \dots, x_k, 0, \dots, 0].$$
 - a. Show that T_k is a linear transformation of V into W_k and that $T_k[V] = \{T_k(\mathbf{v}) \mid \mathbf{v} \text{ in } V\}$ is a subspace of W_k .

- b. If $T_k[V]$ has dimension d_k , show that, for each $j < n$, we have either $d_{j+1} = d_j$ or $d_{j+1} = d_j + 1$.
- c. Assume that A has four columns. Referring to part (b), suppose that $d_1 = d_2 = 1$ and $d_3 = d_4 = 2$. Find the number of pivots in H , and give the location of each.
- d. Repeat part (c) for the case where A has six columns and $d_1 = 1$, $d_2 = d_3 = d_4 = 2$, and $d_5 = d_6 = 3$.
- e. Argue that, for any matrix A , the number of pivots and the location of each pivot in any row-echelon form of A is always the same.
- f. Show that the reduced row-echelon form of a matrix A is unique. [HINT: Consider the nature of the basis for the row space of A given by the nonzero rows of H .]

34. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and let U be a subspace of $\mathbb{R}^n$. Prove that the inverse image $T^{-1}[U]$ is a subspace of $\mathbb{R}^n$.

In Exercises 35–38, let T_1 , T_2 , T_3 , and T_4 be linear transformations whose standard matrix representations are

$$A = \begin{bmatrix} -4 & 5 & 7 \\ 2 & 4 & 5 \\ 1 & 8 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & -3 \\ 1 & 4 \\ 2 & 5 \end{bmatrix},$$

$$C = \begin{bmatrix} -5 & 3 & -6 \\ 11 & 7 & -1 \end{bmatrix}, \quad \text{and}$$

$$D = \begin{bmatrix} 3 & -4 & 1 \\ -2 & 5 & 0 \end{bmatrix},$$

respectively. Use LINTEK or MATLAB to compute the indicated quantity, if it is defined. Load data files for the matrices if the data files are available.

35. $(T_1 \circ T_2 \circ T_4)([1, 2, 1])$
 36. $(T_3 \circ T_1^{-1} \circ T_2)([0, -1])$
 37. $(T_4 \circ (T_2 \circ T_3)^{-1} \circ T_2)([-1, 0])$
 38. $(T_2 \circ (T_3 \circ T_2)^{-1} \circ T_4)([-1, 0, 1])$
 39. Work with Topic 4 of the LINTEK routine VECTGRPH until you can consistently achieve a score of at least 80%.
 40. Work with Topic 5 of the LINTEK routine VECTGRPH until you can regularly attain a score of at least 82%.

2.4

LINEAR TRANSFORMATIONS OF THE PLANE (OPTIONAL)

From the preceding section, we know that every linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by $T(\mathbf{x}) = T_A(\mathbf{x}) = A\mathbf{x}$, where A is some 2×2 matrix. Different 2×2 matrices A give different transformations because $T(\mathbf{e}_1) = A\mathbf{e}_1$ is the first column vector of A and $T(\mathbf{e}_2) = A\mathbf{e}_2$ is the second column vector. The entire plane is mapped onto the column space of the matrix A . In this section we discuss these linear transformations of the plane $\mathbb{R}^2$ into itself, where we can draw reasonable pictures. We will use the familiar x, y -notation for coordinates in the plane.

The Collapsing (Noninvertible) Transformations

For a 2×2 matrix A to be noninvertible, it must have rank 0 or 1. If $\text{rank}(A) = 0$, then A is the zero matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

and we have $A(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^2$. Geometrically, the entire plane is collapsed to a single point—the origin.

If $\text{rank}(A) = 1$, then the column space of A , which is the range of T_A , is a one-dimensional subspace of $\mathbb{R}^2$, which is a line through the origin. The matrix A contains at least one nonzero column vector; if both column vectors are nonzero, then the second one is a scalar multiple of the first one. Examples of such matrices are

$$\begin{array}{cccc} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, & \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, & \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix}, & \begin{bmatrix} 1 & -3 \\ 2 & -6 \end{bmatrix}. & (1) \\ \text{Projection} & \text{Projection} & \text{Collapse} & \text{Collapse} \\ \text{on } x\text{-axis} & \text{on } y\text{-axis} & \text{onto } y = -x & \text{onto } y = 2x \end{array}$$

The first two of these matrices produce *projections* on the coordinate axes, as labeled. *Projection* of the plane on a line L through the origin maps each vector $\mathbf{v}$ onto a vector $\mathbf{p}$ represented geometrically by the arrow starting at the origin and having its tip at the point on L that is closest to the tip of $\mathbf{v}$. The line through the tips of $\mathbf{v}$ and $\mathbf{p}$ must be perpendicular to the line L ; phrased entirely in terms of vectors, the vector $\mathbf{v} - \mathbf{p}$ must be orthogonal to $\mathbf{p}$. This is illustrated in Figure 2.6. Projection on the x -axis is illustrated in Figure 2.7; we see that we have

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

in accord with our labeling of the first matrix in (1). Similarly, the second matrix in (1) gives projection on the y -axis. We refer to such matrices as *projection matrices*. The third and fourth matrices map the plane onto the indicated lines, as we readily see by examination of their column vectors. The transformations represented by these matrices are not projections onto those lines, however. Note that when projecting onto a line, every vector along the

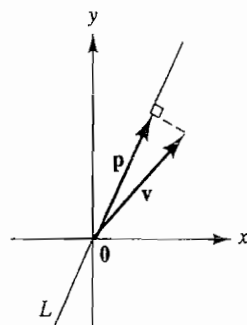


FIGURE 2.6
Projection onto the line L .

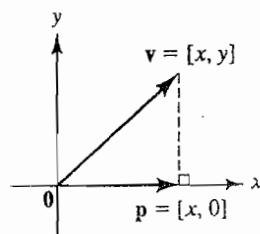


FIGURE 2.7
Projection onto the x -axis.

line is left fixed—that is, it is carried into itself. Now $[3, -3]$ is a vector along the line $y = -x$, but

$$\begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -3 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 3 \\ -3 \end{bmatrix},$$

which shows that the third matrix in (1) is not a projection matrix. A similar computation shows that the final matrix in (1) is not a projection matrix. (See Exercise 1.) Chapter 6 discusses larger projection matrices.

Invertible Linear Transformations of the Plane

We know that if $T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T_A(\mathbf{x}) = A\mathbf{x}$ is an invertible linear transformation of the plane into itself, then A is an invertible 2×2 matrix so that $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^2$. Thus the range of T_A is all of $\mathbb{R}^2$.

Among the invertible linear transformations of the plane are the *rigid motions* of the plane that carry the origin into itself. *Rotation of the plane about the origin* counterclockwise through an angle θ , as illustrated in Figure 2.8, is an example of such a rigid motion.

EXAMPLE 1 Explain geometrically why $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, which rotates the plane counterclockwise through an angle θ , is a linear transformation, and find its standard matrix representation. An algebraic proof is outlined in Exercise 23.

SOLUTION We must show that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ and all scalars r , we have $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(r\mathbf{w}) = rT(\mathbf{w})$. Figure 2.8(a) indicates that the parallelogram that defines $\mathbf{u} + \mathbf{v}$ is carried into the parallelogram defining $T(\mathbf{u}) + T(\mathbf{v})$ by T , and Figure 2.8(b) similarly shows the lines illustrating $r\mathbf{w}$ and $T(r\mathbf{w})$. Thus T preserves addition and scalar multiplication. Figure 2.9 indicates that

$$T(\mathbf{e}_1) = [\cos \theta, \sin \theta] \quad \text{and} \quad T(\mathbf{e}_2) = [-\sin \theta, \cos \theta].$$

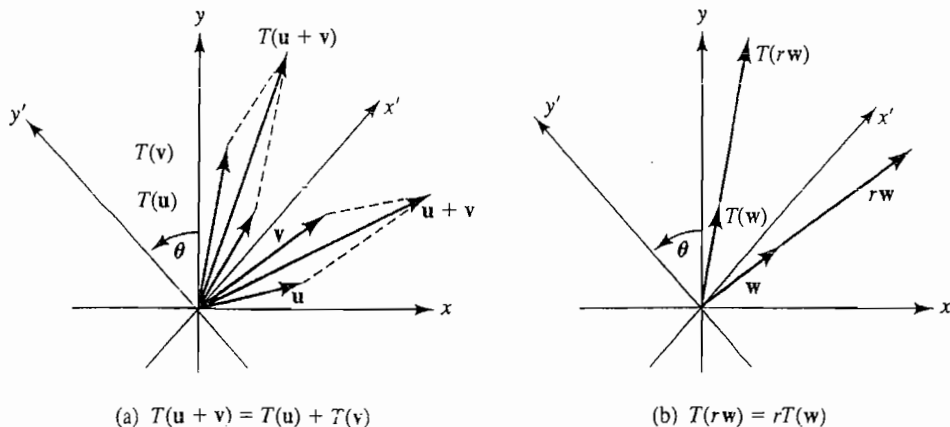


FIGURE 2.8

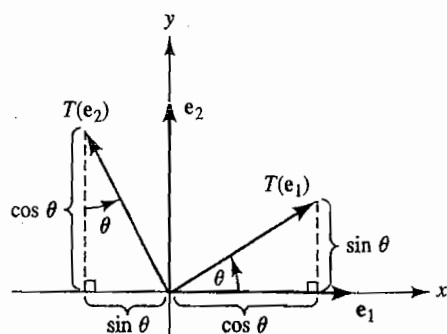


FIGURE 2.9
Counterclockwise rotation of e_1 and e_2
through the angle θ .

Thus

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad \text{Counterclockwise rotation through } \theta \quad (2)$$

is the standard matrix representation of this transformation. ■

Another type of rigid motion T of the plane consists of “turning the plane over” around a line L through the origin. Turn the plane by holding the ends of the line L and rotating 180° , as you might hold a pencil by the ends with the “No. 2” designation on top and rotate it 180° so that the “No. 2” is on the underside. In analogy with the rotation in Figure 2.8, the parallelogram defining $u + v$ is carried into one defining $T(u) + T(v)$, and similarly for the arrows defining the scalar product rw . This type of rigid motion of the plane is called a *reflection in the line L* , because if we think of holding a mirror perpendicular to the plane with its bottom edge falling on L , then $T(v)$ is the reflection of v in this mirror, as indicated in Figure 2.10. Every vector w along L is carried into itself. As indicated in Figure 2.11, the reflection T of the plane

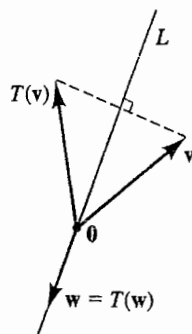


FIGURE 2.10
Reflection in the line L .

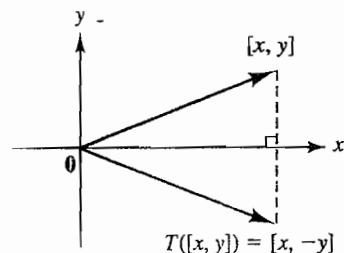


FIGURE 2.11
Reflection in the x -axis.

in the x -axis is defined by $T([x, y]) = [x, -y]$. Because $\mathbf{e}_1$ is left fixed and $\mathbf{e}_2$ is carried into $-\mathbf{e}_2$, we see that the standard matrix representation of this reflection in the x -axis is

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad \text{Reflection in the } x\text{-axis} \quad (3)$$

It can be shown that the rigid motions of the plane carrying the origin into itself are precisely the linear transformations $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that *preserve lengths* of all vectors in $\mathbb{R}^2$ —that is, such that $\|T(\mathbf{x})\| = \|\mathbf{x}\|$ for all $\mathbf{x} \in \mathbb{R}^2$. We will discuss such ideas further in Exercises 17–22.

Thinking for a moment, we can see that every rigid motion of the plane leaving the origin fixed is either a rotation or a reflection followed by a rotation. Namely, if the plane is not turned over, all we can do is rotate it about the origin. If the plane has been turned over, we can achieve its final position by reflection in the x -axis, turning it over horizontally, followed by a rotation about the origin to obtain the desired position. We will use this last fact in the second solution of the next example. (Actually, every rigid motion leaving the origin fixed and turning the plane over is a reflection in some line through the origin, although this is not quite as easy to see.) The first solution of the next example illustrates that bases for $\mathbb{R}^2$ other than the standard basis can be useful.

EXAMPLE 2 Find the standard matrix representation A for the reflection of the plane in the line $y = 2x$.

SOLUTION 1 Let $\mathbf{b}_1 = [1, 2]$, which lies along the line $y = 2x$, and let $\mathbf{b}_2 = [-2, 1]$, which is orthogonal to $\mathbf{b}_1$ because $\mathbf{b}_1 \cdot \mathbf{b}_2 = 0$. These vectors are shown in Figure 2.12. If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is reflection in the line $y = 2x$, then we have

$$T(\mathbf{b}_1) = \mathbf{b}_1 \quad \text{and} \quad T(\mathbf{b}_2) = -\mathbf{b}_2.$$

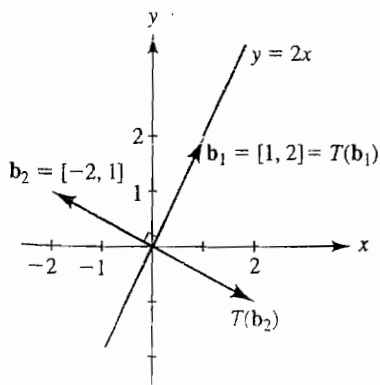


FIGURE 2.12
Reflection in the line $y = 2x$.

Now $\{b_1, b_2\}$ is a basis for $\mathbb{R}^2$, and Theorem 2.7 tells us that T is completely determined by its action on this basis. To find $T(e_1)$ and $T(e_2)$ for the column vectors in the standard matrix representation A of T , we first express e_1 and e_2 as linear combinations of b_1 and b_2 . To do this, we solve the two linear systems with e_1 and e_2 as column vectors of constants and b_1 and b_2 as columns of the coefficient matrix, as follows:

$$\begin{bmatrix} 1 & -2 & | & 1 & 0 \\ 2 & 1 & | & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & | & 1 & 0 \\ 0 & 5 & | & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & | & \frac{1}{5} & \frac{2}{5} \\ 0 & 1 & | & -\frac{2}{5} & \frac{1}{5} \end{bmatrix}.$$

Thus we have

$$e_1 = \frac{1}{5}b_1 - \frac{2}{5}b_2 \quad \text{and} \quad e_2 = \frac{2}{5}b_1 + \frac{1}{5}b_2.$$

Applying the transformation T to both sides of these equations, we obtain

$$T(e_1) = \frac{1}{5}T(b_1) - \frac{2}{5}T(b_2) = \frac{1}{5}b_1 + \frac{2}{5}b_2 = \frac{1}{5}[1, 2] + \frac{2}{5}[-2, 1] = \left[-\frac{3}{5}, \frac{4}{5}\right]$$

and

$$T(e_2) = \frac{2}{5}T(b_1) + \frac{1}{5}T(b_2) = \frac{2}{5}b_1 - \frac{1}{5}b_2 = \frac{2}{5}[1, 2] - \frac{1}{5}[-2, 1] = \left[\frac{4}{5}, \frac{3}{5}\right].$$

Thus the standard matrix representation is

$$A = \begin{bmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}.$$

SOLUTION 2 The three parts of Figure 2.13 show that we can attain the reflection of the plane in the line $y = 2x$ as follows: First reflect in the x -axis, taking us from part (a) to part (b) of the figure, and then rotate counterclockwise through the angle 2θ , where θ is the angle from the x -axis to the line $y = 2x$, measured counterclockwise. Using the double angle formulas derived in Section 2.3, we see from the right triangle in Figure 2.13(a) that

$$\sin 2\theta = 2 \sin \theta \cos \theta = 2 \left(\frac{2}{\sqrt{5}} \right) \left(\frac{1}{\sqrt{5}} \right) = \frac{4}{5}$$

and

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = \frac{1}{5} - \frac{4}{5} = -\frac{3}{5}.$$

Replacing θ by 2θ in the matrix in Example 1, we see that the standard matrix representation for rotation through the angle 2θ is

$$\begin{bmatrix} -\frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix}. \quad (4)$$

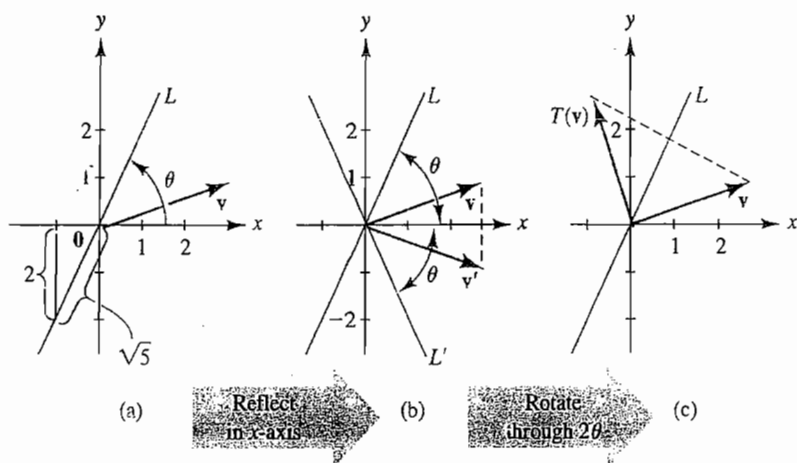


FIGURE 2.13
(a) The vector v (b) Reflected (c) Rotated

Multiplying matrices (4) and (3), we obtain

$$A = \begin{bmatrix} -\frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{3}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}.$$

Rotate Reflect

In Example 2, note that because we first reflect in the x -axis and then rotate through 2θ , the matrix for the reflection is the one on the *right*, which acts on a vector $v \in \mathbb{R}^2$ *first* when computing Av .*

A Geometric Description of All Invertible Transformations of the Plane

We exploit matrix techniques to describe geometrically all invertible linear transformations of the plane into itself. Recall that every invertible matrix is a product of elementary matrices. If we can interpret geometrically the effect on the plane of the linear transformations having elementary 2×2 matrices as their standard representations, we will gain insight into all invertible linear transformations of $\mathbb{R}^2$ into $\mathbb{R}^2$.

*This right-to-left order for composite transformations occurs because we write functions on the left of the elements of the domain on which they act, writing $f(x)$ rather than $(x)f$. From a pedagogical standpoint, writing functions on the left must be regarded as a peculiarity in the development of mathematical notations in a society where text is read from left to right. If we wrote functions on the right side, then we would take the transpose of Ax and write

$$x^T A^T = x^T [\text{reflection matrix}]^T [\text{rotation matrix}]^T,$$

except that we would of course have developed things in terms of row vectors so that the transpose notation would not appear as it does here.

EXAMPLE 3 Describe geometrically the effect on the plane of the linear transformation T_E where E is an elementary matrix obtained by multiplying a row of the 2×2 identity matrix I by -1 .

SOLUTION The matrix obtained by multiplying the second row of I by -1 is

$$E = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix},$$

which is the matrix given as matrix (3). We saw there that T_E is the reflection in the x -axis and $T_E([x, y]) = [x, -y]$. Similarly, we see that the elementary matrix obtained by multiplying the first row of I by -1 represents the transformation that changes the sign of the first component of a vector, carrying $[x, y]$ into $[-x, y]$. This is the reflection in the y -axis. ■

EXAMPLE 4 Describe geometrically the effect on the plane of the linear transformation T_E where E is an elementary matrix obtained by interchanging the rows of the 2×2 identity matrix I .

SOLUTION Here we have

$$E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}.$$

In row-vector notation, we have $T_E([x, y]) = [y, x]$. Figure 2.14 indicates that this transformation, which interchanges the components of a vector in the plane, is the reflection in the line $y = x$. ■

EXAMPLE 5 Describe geometrically the linear transformation $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = E\begin{bmatrix} x \\ y \end{bmatrix}$, where E is a 2×2 elementary matrix corresponding to row scaling.

SOLUTION The matrix E has the form

$$\begin{bmatrix} r & 0 \\ 0 & 1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 & 0 \\ 0 & r \end{bmatrix}$$

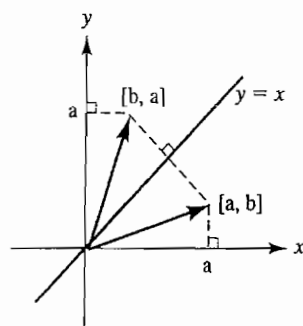


FIGURE 2.14
Reflection in the line $y = x$.

for some nonzero scalar r . We discuss the first case and leave the second as Exercise 8. The transformation is given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} r & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} rx \\ y \end{bmatrix},$$

or, in row notation, $T([x, y]) = [rx, y]$. The second component of $[x, y]$ is unchanged. However, the first component is multiplied by the scalar r , resulting in a horizontal expansion if $r > 1$ or in a horizontal contraction if $0 < r < 1$. In Figure 2.15, we illustrate the effect of such a horizontal expansion or contraction on the points of the unit circle. If $r < 0$, we have an expansion or contraction followed by a reflection in the y -axis. For example,

$$\begin{bmatrix} -3 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix},$$

Reflection Horizontal expansion

indicating a horizontal expansion by a factor of 3, followed by a reflection in the y -axis. ■

EXAMPLE 6 Describe geometrically the linear transformation $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = E\begin{bmatrix} x \\ y \end{bmatrix}$, where E is a 2×2 elementary matrix corresponding to row addition.

SOLUTION The matrix E has the form

$$\begin{bmatrix} 1 & 0 \\ r & 1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 & r \\ 0 & 1 \end{bmatrix}$$

for some nonzero scalar r . We discuss the first case, and leave the second as Exercise 10. The transformation is given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ r & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ rx + y \end{bmatrix},$$

or, in row-vector notation, $T([x, y])$ and $[x, rx + y]$. The first component of the vector $[x, y]$ is unchanged. However, the second component is changed by the

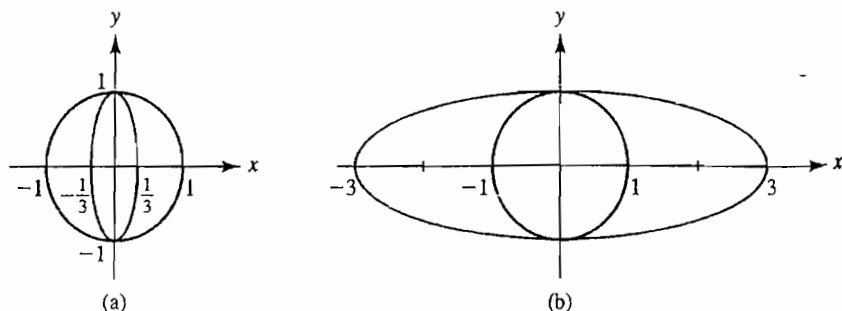


FIGURE 2.15

(a) $T([x, y]) = \begin{bmatrix} 1/3 x \\ y \end{bmatrix}$ contracts horizontally; (b) $T([x, y]) = [3x, y]$ expands horizontally.

addition of rx . For example, $[1, 0]$ is carried onto $[1, r]$, and $[1, 1]$ is carried onto $[1, 1 + r]$, while $[0, 0]$ and $[0, 1]$ are carried onto themselves. Notice that every vector along the y -axis remains fixed. Figure 2.16 illustrates the effect of this transformation. The squares shaded in black are carried onto the parallelograms shaded in color. This transformation is called a *vertical shear*. Exercise 10 deals with the case of a horizontal shear. ■

We have noted that a square matrix A is invertible if and only if it is the product of elementary matrices. We also know that a product of matrices corresponds to the composition of the associated linear transformations, and we have seen the effect of transformations associated with elementary matrices on the plane. Putting all of these ideas together, we obtain the following.

Geometric Description of Invertible Transformations of $\mathbb{R}^2$

A linear transformation T of the plane $\mathbb{R}^2$ into itself is invertible if and only if T consists of a finite sequence of:

- Reflections in the x -axis, the y -axis, or the line $y = x$;
- Vertical or horizontal expansions or contractions; and
- Vertical or horizontal shears.

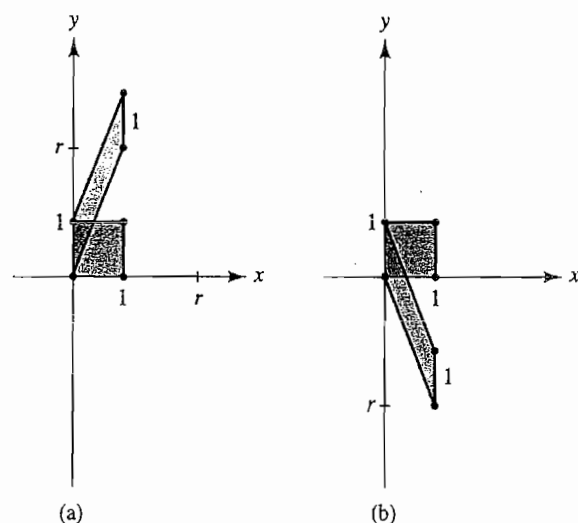


FIGURE 2.16

(a) The vertical shear $T([x, y]) = [x, rx + y]$, $r > 0$

(b) the vertical shear $T([x, y]) = [x, rx + y]$, $r < 0$

EXAMPLE 7 Illustrate the result relating to the boxed description above for the invertible linear transformation $T([x, y]) = [x + 2y, 3x + 4y]$.

SOLUTION We reduce the standard matrix representation A of T , obtaining

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow[R_2 \rightarrow R_2 - 3R_1]{E_1} \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix} \xrightarrow[R_2 \rightarrow -\frac{1}{2}R_2]{E_2} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \xrightarrow[R_1 \rightarrow R_1 - 2R_2]{E_3} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

In terms of elementary matrices, this reduction becomes

$$\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} A = I,$$

$E_3 \qquad E_2 \qquad E_1$

and so

$$A = E_1^{-1} E_2^{-1} E_3^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

This shows that T consists of a horizontal shear (matrix E_3^{-1}) followed by an expansion and reflection (matrix E_2^{-1}), followed in turn by a vertical shear (matrix E_1^{-1}). ■

SUMMARY

1. Linear transformations of $\mathbb{R}^2$ into $\mathbb{R}^2$ whose standard matrix representations have rank less than 2 either collapse the entire plane to the origin (the rank 0 case) or collapse the plane to a line (the rank 1 case).
2. A rigid motion of the plane into itself that leaves the origin fixed gives a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Every such rigid motion is either a rotation of the plane about the origin, or a reflection in the x -axis (to turn the plane over) followed by such a rotation.
3. The standard matrix representation for the rotation of the plane counter-clockwise about the origin through an angle θ is

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

4. An invertible linear transformation of $\mathbb{R}^2$ into itself can be described geometrically, using elementary matrices, as indicated in the box preceding Example 7.

EXERCISES

1. Explain why the linear transformation $T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, where $A = \begin{bmatrix} 1 & -3 \\ 2 & -6 \end{bmatrix}$, has the line $y = 2x$ as range, but is not the projection of $\mathbb{R}^2$ onto that line.
2. Give the standard matrix representation of the rotation of the plane *counterclockwise* about the origin through an angle of
 - a. 30° ,
 - b. 90° ,
 - c. 135° .
3. Give the standard matrix representation of the rotation of the plane *clockwise* about the origin through an angle of
 - a. 45° ,
 - b. 60° ,
 - c. 150° .
4. Use the rotation matrix in item 3 of the Summary to derive trigonometric identities for $\sin 3\theta$ and $\cos 3\theta$ in terms of $\sin \theta$ and $\cos \theta$. (See Illustration 1, Section 2.3.)
5. Use the rotation matrix in item 3 of the Summary to derive trigonometric identities for $\sin(\theta + \phi)$ and $\cos(\theta + \phi)$ in terms of $\sin \theta$, $\sin \phi$, $\cos \theta$, and $\cos \phi$. (See Illustration 1, Section 2.3.)
6. Find the general matrix representation for the reflection of the plane in the line $y = mx$, using the method for the case $m = 2$ in Solution 1 of Example 2 in the text.
7. Repeat Exercise 6, but use the method for the case $m = 2$ in Solution 2 of Example 2 in the text.
8. Show that the linear transformation

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

affects the plane $\mathbb{R}^2$ as follows:

- (i) A vertical expansion, if $r > 1$;
- (ii) A vertical contraction, if $0 < r < 1$;
- (iii) A vertical expansion followed by a reflection in the x -axis, if $r < -1$;
- (iv) A vertical contraction followed by a reflection in the x -axis, if $-1 < r < 0$.

9. Referring to Exercise 8, explain algebraically why cases (iii) and (iv) can be described by the reflection followed by the expansion or contraction, in that order.

10. Show that the linear transformation

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & r \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

corresponds to a horizontal shear of the plane.

In Exercises 11–15, express the standard matrix representation of the given invertible transformation of $\mathbb{R}^2$ into itself as a product of elementary matrices. Use this expression to describe the transformation as a product of one or more reflections, horizontal or vertical expansions or contractions, and shears.

11. $T([x, y]) = [-y, x]$ (Rotation counterclockwise through 90°)
12. $T([x, y]) = [2x, 2y]$ (Expansion away from the origin by a factor of 2)
13. $T([x, y]) = [-x, -y]$ (Rotation through 180°)
14. $T([x, y]) = [x + y, 2x - y]$
15. $T([x, y]) = [x + y, 3x + 5y]$
16. Mark each of the following True or False.
 - a. Every rotation of the plane is a linear transformation.
 - b. Every rotation of the plane about the origin is a linear transformation.
 - c. Every reflection of the plane in a line L is a rigid motion of the plane.
 - d. Every reflection of the plane in a line L is a linear transformation of the plane.
 - e. Every rigid motion of the plane that carries the origin into itself is a linear transformation.
 - f. Every invertible linear transformation of the plane is a rigid motion.
 - g. If a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a rigid motion of the plane, then $\|T(x)\| = \|x\|$ for all $x \in \mathbb{R}^2$.

- h. The geometric effect of all invertible linear transformations of $\mathbb{R}^3$ into itself can be described in terms of the geometric effect of the linear transformations of $\mathbb{R}^3$ having elementary matrices as standard matrix representations.
- i. Every linear transformation of the plane into itself can be achieved through a succession of reflections, expansions, contractions, and shears.
- j. Every invertible linear transformation of the plane into itself can be achieved through a succession of reflections, expansions, contractions, and shears.

A linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ preserves length if $\|T(\mathbf{x})\| = \|\mathbf{x}\|$ for all $\mathbf{x} \in \mathbb{R}^2$. It preserves angle if the angle between $\mathbf{u}$ and $\mathbf{v}$ is the same as the angle between $T(\mathbf{u})$ and $T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$. It preserves the dot product if $T(\mathbf{u}) \cdot T(\mathbf{v}) = \mathbf{u} \cdot \mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$.

We recommend that Exercises 17–22 be worked sequentially, or at least be read sequentially.

17. Use the familiar equation that describes the dot product $\mathbf{u} \cdot \mathbf{v}$ geometrically to prove that if a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ preserves both length and angle, then it also preserves the dot product.
18. Use algebraic properties of the dot product to compute $\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})$, and prove from the resulting equation that a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that preserves length also preserves the dot product.
19. Express both the length of a vector $\mathbf{v} \in \mathbb{R}^2$ and the angle between two nonzero vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ in terms of the dot product only. (From this we may conclude that if a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ preserves the dot product, then it preserves length and angle.)
20. Suppose that $T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ preserves both length and angle. Prove that the two column vectors of the matrix A are orthogonal unit vectors.
21. Prove that the two column vectors of a 2×2 matrix A are orthogonal unit vectors if and only if $(A^T)A = I$. Demonstrate that the matrix representations for the rigid motions given in Examples 1 and 2 satisfy this condition.
22. Let A be a 2×2 matrix such that $(A^T)A = I$. Prove that the linear transformation T_A preserves the dot product, and hence also preserves length and angle. [HINT: Note that the dot product of two column vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is the entry in the 1×1 matrix $(\mathbf{u}^T)\mathbf{v}$. Compute the dot product $T_A(\mathbf{u}) \cdot T_A(\mathbf{v})$ by computing $(A\mathbf{u})^T(A\mathbf{v})$.]
23. This exercise outlines an algebraic proof that rotation of the plane about the origin is a linear transformation. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the function that rotates the plane counterclockwise through an angle θ as in Example 1.
- Prove algebraically that each vector $\mathbf{v} \in \mathbb{R}^2$ can be written in the polar form $\mathbf{v} = r[\cos \alpha, \sin \alpha]$. [HINT: Each unit vector has this form with $r = 1$.]
 - For $\mathbf{v} = r[\cos \alpha, \sin \alpha]$, express $T(\mathbf{v})$ in this polar form.
 - Using column-vector notation and appropriate trigonometric identities, find a matrix A such that $T(\mathbf{v}) = A\mathbf{v}$. The existence of such a matrix A proves that T is a linear transformation.

2.5

LINES, PLANES, AND OTHER FLATS (Optional)

We turn to geometry in this section, and generalize the notions of a *line* in the plane or in space and of a *plane* in space. Our work in the preceding sections will enable us to describe geometrically the solution set of any consistent linear system.

The Notion of a Line in $\mathbb{R}^n$

For a nonzero vector $\mathbf{d}$ in $\mathbb{R}^2$, we visualize the one-dimensional subspace $\text{sp}(\mathbf{d})$ as a line through the origin, as shown in Figure 2.17. Similarly, Figure 2.18 indicates that for a nonzero vector $\mathbf{d}$ in $\mathbb{R}^n$, we can view $\text{sp}(\mathbf{d}) = \{t\mathbf{d} \mid t \in \mathbb{R}\}$ as a line through the origin. Every subspace of $\mathbb{R}^n$ contains the origin (zero vector), but we surely want to consider lines in $\mathbb{R}^n$ that do not pass through the origin, such as line L in Figure 2.19. As indicated in Figure 2.19, if $\mathbf{a}$ is a vector to a point on the line L , then every point on L is at the tip of a vector $\mathbf{x} = t\mathbf{d} + \mathbf{a}$, where t is a scalar and $\mathbf{d}$ is any fixed nonzero vector that we regard intuitively as *parallel* to the line L . This line is thus obtained from the line $\text{sp}(\mathbf{d})$ by *translation*. Geometers consider a translation of a subset S of $\mathbb{R}^n$ to be a sliding of every point in S in the same direction and for the same distance. The direction and distance for a translation can be specified by a vector (such as vector $\mathbf{a}$ above) pointing in the direction of the translation and having magnitude equal to the distance the points are moved. The image of S under such a translation is a *translate* of S . We will give a formal definition of a translate of a subset, present an example, and then define a line in $\mathbb{R}^n$.

DEFINITION 2.4 Translate of a Subset of $\mathbb{R}^n$

Let S be a subset of $\mathbb{R}^n$ and let $\mathbf{a}$ be a vector in $\mathbb{R}^n$. The set $\{\mathbf{x} + \mathbf{a} \mid \mathbf{x} \in S\}$ is the *translate* of S by $\mathbf{a}$, and is denoted by $S + \mathbf{a}$. The vector $\mathbf{a}$ is the *translation vector*.

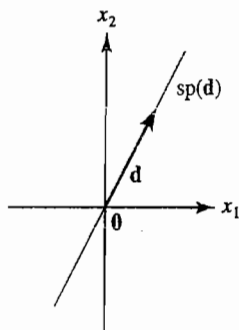


FIGURE 2.17
A line through the origin in $\mathbb{R}^2$.

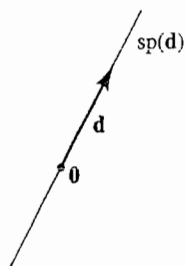


FIGURE 2.18
A line through the origin in $\mathbb{R}^n$.

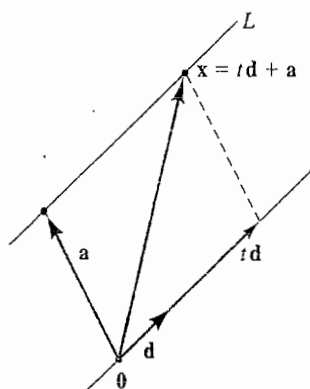


FIGURE 2.19
A general line L in $\mathbb{R}^n$.

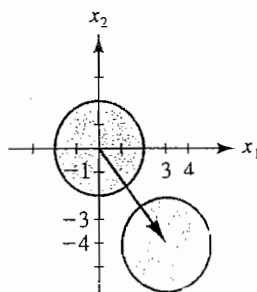


FIGURE 2.20
Translate of $\{x \in \mathbb{R}^2 \mid \|x\| \leq 2\}$ in $\mathbb{R}^2$
by $[3, -4]$.

EXAMPLE 1 Sketch the translate of the subset $S = \{x \in \mathbb{R}^2 \mid \|x\| \leq 2\}$ of $\mathbb{R}^2$ by the vector $[3, -4]$.

SOLUTION The subset S of $\mathbb{R}^2$ is a disk with center at the origin and a radius of 2. As shown in Figure 2.20, its translate by $[3, -4]$ is the disk with center at the point $(3, -4)$ and a radius of 2. ■

DEFINITION 2.5 Line in $\mathbb{R}^n$

A **line** in $\mathbb{R}^n$ is a translate of a one-dimensional subspace of $\mathbb{R}^n$.

Although our definition defines a line to be a set of vectors, it is customary in geometry to consider a line as the set of *points* in $\mathbb{R}^n$ whose coordinates correspond to the components of the vectors.

We can specify a line L in $\mathbb{R}^n$ by giving a point $(a_1, a_2, \dots, a_n)$ on the line and a vector $\mathbf{d}$ parallel to the line. We consider $\mathbf{d}$ to be a *direction vector* for the line, whereas $\mathbf{a}$ is a *translation vector*. In the terminology of Definition 2.5, L is the translate of the subspace $\text{sp}(\mathbf{d})$ by the vector $\mathbf{a} = [a_1, a_2, \dots, a_n]$ —that is, $L = \{t\mathbf{d} + \mathbf{a} \mid t \in \mathbb{R}\}$. We can describe the line by the single equation

$$\mathbf{x} = t\mathbf{d} + \mathbf{a} \quad \text{Vector equation of } L$$

or by the equations

$$x_1 = td_1 + a_1$$

$$x_2 = td_2 + a_2$$

$$\vdots$$

$$\vdots$$

$$x_n = td_n + a_n.$$

Component equations of L

In classical geometry, component equations for L are also called **parametric equations** for L , and the variable t is a **parameter**. Of course, this same parameter t appears in the vector equation also.

EXAMPLE 2 Find a vector equation and component equations for the line in $\mathbb{R}^2$ through $(2, 1)$ having direction vector $[3, 4]$. Then find the point on the line having -4 as its x_1 -coordinate.

SOLUTION The line can be characterized as the translate of $\text{sp}([3, 4])$ by the vector $[2, 1]$, and so a vector equation of the line is

$$[x_1, x_2] = t[3, 4] + [2, 1].$$

The component equations are

$$x_1 = 3t + 2, \quad x_2 = 4t + 1.$$

Because t runs through all real numbers, we obtain all points (x_1, x_2) on the line from these component equations. In order to find the point on the line with -4 as x_1 -coordinate, we set $-4 = 3t + 2$ and obtain $t = -2$. Thus the x_2 -coordinate is $4(-2) + 1 = -7$, and so the desired point is $(-4, -7)$. ■

EXAMPLE 3 Find parametric equations of the line in $\mathbb{R}^3$ that passes through the points $(2, -1, 3)$ and $(1, 3, 5)$.

SOLUTION We arbitrarily choose $\mathbf{a} = [2, -1, 3]$ as the translation vector corresponding to the point $(2, -1, 3)$ on the line. A direction vector is given by

$$\mathbf{d} = [1, 3, 5] - [2, -1, 3] = [-1, 4, 2],$$

as indicated in Figure 2.21. We obtain

$$[x_1, x_2, x_3] = t[-1, 4, 2] + [2, -1, 3]$$

as a vector equation for the line. The corresponding parametric (component) equations are

$$x_1 = -t + 2, \quad x_2 = 4t - 1, \quad x_3 = 2t + 3. \quad \blacksquare$$

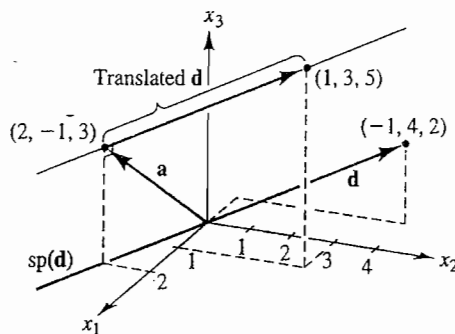


FIGURE 2.21

The line passing through $(2, -1, 3)$ and $(1, 3, 5)$.

Line Segments

Consider the line in $\mathbb{R}^n$ that passes through the two points $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$. Letting $\mathbf{a}$ and $\mathbf{b}$ be the vectors corresponding to these points, we see as in Example 3 that $\mathbf{d} = \mathbf{b} - \mathbf{a}$ is a direction vector for the line. The vector equation

$$\mathbf{x} = t\mathbf{d} + \mathbf{a} = t(\mathbf{b} - \mathbf{a}) + \mathbf{a} \quad (1)$$

for the line in effect presents the line as a t -axis whose origin is at the point $(a_1, a_2, \dots, a_n)$ and on which a one-unit change in t corresponds to $\|\mathbf{d}\|$ units distance in $\mathbb{R}^n$. This is illustrated in Figure 2.22.

As illustrated in Figure 2.23, each point in $\mathbb{R}^n$ on the line segment that joins the tip of $\mathbf{a}$ to the tip of $\mathbf{b}$ lies at the tip of a vector $\mathbf{x}$ obtained in Eq. (1) for some value of t for which $0 \leq t \leq 1$. Note that $t = 0$ yields the point at the tip of $\mathbf{a}$ and $t = 1$ yields the point at the tip of $\mathbf{b}$. By choosing t between 0 and 1 appropriately, we can find the coordinates of any point on this line segment. In particular, the coordinates of the **midpoint** of the line segment are the components of the vector

$$\mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}(\mathbf{a} + \mathbf{b}).$$

EXAMPLE 4 Find the points that divide into five equal parts the line segment that joins $(1, 2, 1, 3)$ to $(2, 1, 4, 2)$ in $\mathbb{R}^4$.

SOLUTION We obtain $\mathbf{d} = [2, 1, 4, 2] - [1, 2, 1, 3] = [1, -1, 3, -1]$ as a direction vector for the line through the two given points. The corresponding vector equation of the line is

$$[x_1, x_2, x_3, x_4] = t[1, -1, 3, -1] + [1, 2, 1, 3].$$

By choosing $t = 0, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}$, and 1, we obtain coordinates of the points that divide the segment as required, as shown in Table 2.1. ■

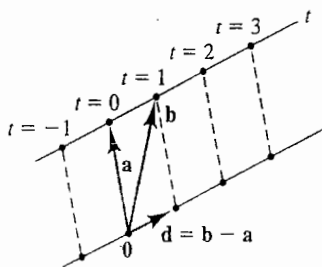


FIGURE 2.22
Equation (1) sets up a t -axis.

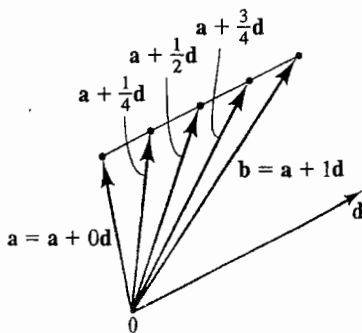


FIGURE 2.23
Points on a line segment.

TABLE 2.1

t	Equally Spaced Points
0	(1, 2, 1, 3)
$\frac{1}{5}$	(1.2, 1.8, 1.6, 2.8)
$\frac{2}{5}$	(1.4, 1.6, 2.2, 2.6)
$\frac{3}{5}$	(1.6, 1.4, 2.8, 2.4)
$\frac{4}{5}$	(1.8, 1.2, 3.4, 2.2)
1	(2, 1, 4, 2)

Flats in $\mathbb{R}^n$

Just as a line is a translate of a one-dimensional subspace in $\mathbb{R}^n$, a *plane* in $\mathbb{R}^n$ is a translate of a two-dimensional subspace $\text{sp}(\mathbf{d}_1, \mathbf{d}_2)$, where $\mathbf{d}_1$ and $\mathbf{d}_2$ are nonzero, nonparallel vectors in $\mathbb{R}^n$. A plane appears as a *flat* piece of $\mathbb{R}^n$, as illustrated in Figure 2.24. We have no word analogous to “straight” or “flat” in our language to denote that $\mathbb{R}^3$ is not “curved.” We borrow the term “flat” when generalizing to higher dimensions, and describe a translate of a k -dimensional subspace of $\mathbb{R}^n$ for $k < n$ as being “flat.” Let us give a formal definition.

DEFINITION 2.6 A k -Flat in $\mathbb{R}^n$

A k -flat in $\mathbb{R}^n$ is a translate of a k -dimensional subspace of $\mathbb{R}^n$. In particular, a 1-flat is a **line**, a 2-flat is a **plane**, and an $(n - 1)$ -flat is a **hyperplane**. We consider each point of $\mathbb{R}^n$ to be a zero-flat.

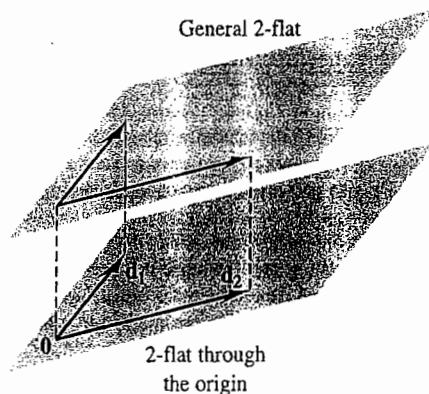


FIGURE 2.24
Planes or 2-flats in $\mathbb{R}^n$

Just as for a line, it is conventional in geometry to speak of the translate $W + \mathbf{a}$ of a k -dimensional subspace W of $\mathbb{R}^n$ as the k -flat through the point $(a_1, a_2, \dots, a_n)$ parallel to W . If $\{\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_k\}$ is a basis for W , then

$$\mathbf{x} = t_1 \mathbf{d}_1 + t_2 \mathbf{d}_2 + \dots + t_k \mathbf{d}_k + \mathbf{a} \quad (2)$$

is the **vector equation** of the k -flat. (We use the letter $\mathbf{d}$ because W determines the direction of the k -flat as being parallel to W .) The corresponding component equations are again called **parametric equations** for the k -flat.

EXAMPLE 5 Find parametric equations of the plane in $\mathbb{R}^4$ passing through the points $(1, 1, 1, 1)$, $(2, 1, 1, 0)$, and $(3, 2, 1, 0)$.

SOLUTION We arbitrarily choose $\mathbf{a} = [1, 1, 1, 1]$ as the translation vector corresponding to the point $(1, 1, 1, 1)$ on the desired plane. Two vectors that (when translated) start at this point and reach to the other two points are

$$\mathbf{d}_1 = [2, 1, 1, 0] - [1, 1, 1, 1] = [1, 0, 0, -1]$$

and

$$\mathbf{d}_2 = [3, 2, 1, 0] - [1, 1, 1, 1] = [2, 1, 0, -1].$$

Because these vectors are nonparallel, they form a basis for the 2-flat through the origin and parallel to the desired plane. See Figure 2.25. The vector equation of the plane is $\mathbf{x} = s\mathbf{d}_1 + t\mathbf{d}_2 + \mathbf{a}$, or, written out,

$$[x_1, x_2, x_3, x_4] = s[1, 0, 0, -1] + t[2, 1, 0, -1] + [1, 1, 1, 1].$$

HISTORICAL NOTE THE EQUATION OF A PLANE IN $\mathbb{R}^3$ appears as early as 1732 in a paper of Jacob Hermann (1678–1733). He was able to determine the plane's position by using intercepts, and he also noted that the sine of the angle between the plane and the one coordinate plane he dealt with (what we call the x_1, x_2 -plane) was

$$\frac{\sqrt{d_1^2 + d_2^2}}{\sqrt{d_1^2 + d_2^2 + d_3^2}}$$

In his 1748 *Introduction to Infinitesimal Analysis*, Leonhard Euler (1707–1783) used, instead, the cosine of this angle, $d_3/\sqrt{d_1^2 + d_2^2 + d_3^2}$.

At the end of the eighteenth century, Gaspard Monge (1746–1818), in his notes for a course on solid analytic geometry at the Ecole Polytechnique, related the equation of a plane to all three coordinate planes and gave the cosines of the angles the plane made with each of these (the so-called direction cosines). He also presented many of the standard problems of solid analytic geometry, examples of which appear in the exercises. For instance, he showed how to find the plane passing through three given points, the line passing through a point perpendicular to a plane, the distance between two parallel planes, and the angle between a line and a plane.

Known as “the greatest geometer of the eighteenth century,” Monge developed new graphical geometric techniques as a student and later as a professor at a military school. The first problem he solved had to do with a procedure enabling soldiers to make quickly a fortification capable of shielding a position from both the view and the firepower of the enemy. Monge served the French revolutionary government as minister of the navy and later served Napoleon in various scientific offices. Ultimately, he was appointed senator for life by the emperor.

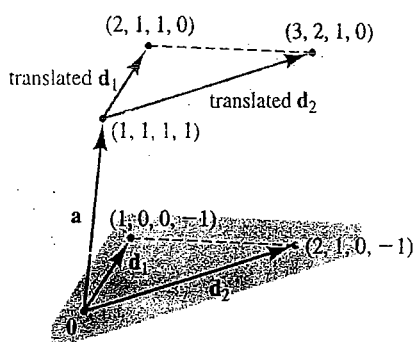


FIGURE 2.25
A 2-flat in $\mathbb{R}^4$.

Parametric equations obtained by equating components are

$$\begin{aligned}x_1 &= s + 2t + 1 \\x_2 &= t + 1 \\x_3 &= 1 \\x_4 &= -s - t + 1.\end{aligned}$$

The Geometry of Linear Systems

Let $Ax = b$ be any system of m equations in n unknowns that has at least one solution $x = p$. Theorem 1.18 on p. 97 shows that the solution set of the system consists of all vectors of the form $x = p + h$, where h is a solution of the homogeneous system $Ax = 0$. Because the solution set of $Ax = 0$ is a subspace of $\mathbb{R}^n$, we see that the solution set of $Ax = b$ is the translate of this subspace by the vector p . That is, the solution set of $Ax = b$ is a k -flat, where k is the nullity of A . If the system of equations has a unique solution, its solution set is a zero-flat.

EXAMPLE 6 Show that the linear equation $c_1x_1 + c_2x_2 + c_3x_3 = b$, where at least one of c_1, c_2, c_3 is nonzero, represents a plane in $\mathbb{R}^3$.

SOLUTION Let us assume that $c_1 \neq 0$. A particular solution of the given equation is $a = [b/c_1, 0, 0]$. The corresponding homogeneous equation has a solution space generated by $d_1 = [c_3, 0, -c_1]$ and $d_2 = [c_2, -c_1, 0]$. Thus the solution set of the linear equation is a 2-flat in $\mathbb{R}^3$ with equation $x = sd_1 + td_2 + a$ —that is, a plane in $\mathbb{R}^3$. ■

Reasoning as in Example 6, we see that every linear equation

$$c_1x_1 + c_2x_2 + \cdots + c_nx_n = b$$

represents a hyperplane—that is, an $(n - 1)$ -flat in $\mathbb{R}^n$.

EXAMPLE 7 Solve the system of equations

$$\begin{aligned}x_1 + 2x_2 - 2x_3 + x_4 + 3x_5 &= 1 \\2x_1 + 5x_2 - 3x_3 - x_4 + 2x_5 &= 2 \\-3x_1 - 8x_2 + 6x_3 - x_4 - 5x_5 &= 1,\end{aligned}$$

and write the solution set as a k -flat.

SOLUTION Reducing the corresponding partitioned matrix, we have

$$\begin{aligned}&\left[\begin{array}{ccccc|c}1 & 2 & -2 & 1 & 3 & 1 \\2 & 5 & -3 & -1 & 2 & 2 \\-3 & -8 & 6 & -1 & -5 & 1\end{array}\right] \sim \left[\begin{array}{ccccc|c}1 & 2 & -2 & 1 & 3 & 1 \\0 & 1 & 1 & -3 & -4 & 0 \\0 & -2 & 0 & 2 & 4 & 4\end{array}\right] \\&\sim \left[\begin{array}{ccccc|c}1 & 0 & -4 & 7 & 11 & 1 \\0 & 1 & 1 & -3 & -4 & 0 \\0 & 0 & 2 & -4 & -4 & 4\end{array}\right] \sim \left[\begin{array}{ccccc|c}1 & 0 & 0 & -1 & 3 & 9 \\0 & 1 & 0 & -1 & -2 & -2 \\0 & 0 & 1 & -2 & -2 & 2\end{array}\right].\end{aligned}$$

Thus, $\mathbf{a} = [9, -2, 2, 0, 0]$ is a particular solution to the given system, and $\mathbf{d}_1 = [1, 1, 2, 1, 0]$ and $\mathbf{d}_2 = [-3, 2, 2, 0, 1]$ form a basis for the solution space of the corresponding homogeneous system. The solution set of the given system is the 2-flat in $\mathbb{R}^5$ with vector equation $\mathbf{x} = \mathbf{a} + t_1\mathbf{d}_1 + t_2\mathbf{d}_2$, which can be written in the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 9 \\ -2 \\ 2 \\ 0 \\ 0 \end{bmatrix} + t_1 \begin{bmatrix} 1 \\ 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} -3 \\ 2 \\ 2 \\ 0 \\ 1 \end{bmatrix}.$$

In the preceding example, we described the solution set of a system of equations as a 2-flat in $\mathbb{R}^5$. Notice that the original form of the system represents an intersection of three hyperplanes in $\mathbb{R}^5$ —one for each equation. We generalize this example to the solution set of any consistent linear system.

Consider a system of m equations in n unknowns. Let the rank of the coefficient matrix be r so that, when the matrix is reduced to row-echelon form, there are r nonzero rows. According to the rank equation, the number of free variables is then $n - r$; the corresponding homogeneous system has as its solution set an $(n - r)$ -dimensional subspace—that is, an $(n - r)$ -flat through the origin. The solution set of the original nonhomogeneous system is a translate of this subspace and is an $(n - r)$ -flat in $\mathbb{R}^n$. In particular, a single consistent linear equation has as its solution set an $(n - 1)$ -flat in $\mathbb{R}^n$. In general, if we adjoin an additional linear equation to a given linear system, we expect the dimension of the solution flat to be reduced by 1. This is the case precisely when the new system is still consistent and when the new equation is independent of the others (in the sense that it yields a new nonzero row when the augmented matrix is row-reduced to echelon form).

We have shown that a system $A\mathbf{x} = \mathbf{b}$ of m equations in n unknowns has as its solution set an $(n - r)$ -flat, where r is the rank of A . Conversely, it can be

shown that a k -flat in $\mathbb{R}^n$ is the solution set of some system of $n - k$ linear equations in n unknowns. That is, a k -flat in $\mathbb{R}^n$ is the intersection of $n - k$ hyperplanes. Thus there are two ways to view a k -flat in $\mathbb{R}^n$:

1. As a translate of a k -dimensional subspace of $\mathbb{R}^n$, described using parametric equations
2. As an intersection of $n - k$ hyperplanes, described with a system of linear equations.

EXAMPLE 8 Describe the line (1-flat) in $\mathbb{R}^3$ that passes through $(2, -1, 3)$ and $(1, 3, 5)$ in terms of

- (1) parametric equations, and
- (2) a system of linear equations.

SOLUTION (1) In Example 3, we found the parametric equations for the line:

$$x_1 = -t + 2, \quad x_2 = 4t - 1, \quad x_3 = 2t + 3. \quad (3)$$

(2) In order to describe the line with a system of linear equations, we eliminate the parameter t from Eqs. (3):

$$\begin{aligned} 4x_1 + x_2 &= 7 && \text{Add four times the first to the second.} \\ x_2 - 2x_3 &= -7 && \text{Subtract twice the third from the second.} \end{aligned} \quad (4)$$

This system describes the line as an intersection of two planes. The line can be represented as the intersection of any two distinct planes, each containing the line. This is illustrated by the equivalent systems we have at the various stages in the Gauss reduction of system (4) to obtain solution (3). ■

SUMMARY

1. The translate of a subset S of $\mathbb{R}^n$ by a vector $\mathbf{a} \in \mathbb{R}^n$ is the set of all vectors in $\mathbb{R}^n$ of the form $\mathbf{x} + \mathbf{a}$ for $\mathbf{x} \in S$, and is denoted by $S + \mathbf{a}$.
2. A k -flat in $\mathbb{R}^n$ is a translate of a k -dimensional subspace and has the form $\mathbf{a} + \text{sp}(\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_k)$, where $\mathbf{a}$ is a vector in $\mathbb{R}^n$ and $\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_k$ are independent vectors in $\mathbb{R}^n$. The vector equation of the k -flat is $\mathbf{x} = \mathbf{a} + t_1\mathbf{d}_1 + t_2\mathbf{d}_2 + \dots + t_k\mathbf{d}_k$ for scalars t_i in $\mathbb{R}$.
3. A line in $\mathbb{R}^n$ is a 1-flat. The line passing through the point $\mathbf{a}$ with parallel vector $\mathbf{d}$ is given by $\mathbf{x} = \mathbf{a} + t\mathbf{d}$, where t runs through all scalars. Parametric equations of the line are the component equations $x_i = a_i + d_i t$ for $i = 1, 2, \dots, n$.
4. Let $\mathbf{a}$ and $\mathbf{b}$ be vectors in $\mathbb{R}^n$. Vectors to points on the line segment from the tip of $\mathbf{a}$ to the tip of $\mathbf{b}$ are vectors of the form $\mathbf{x} = t(\mathbf{b} - \mathbf{a}) + \mathbf{a}$ for $0 \leq t \leq 1$.
5. A plane in $\mathbb{R}^n$ is a 2-flat; a hyperplane in $\mathbb{R}^n$ is an $(n - 1)$ -flat.

6. The solution set of a consistent linear system in n variables with coefficient matrix of rank r is an $(n - r)$ -flat in $\mathbb{R}^n$.
7. Every k -flat in $\mathbb{R}^n$ can be viewed both as a translate of a k -dimensional subspace and as the intersection of $n - k$ hyperplanes.

EXERCISES

In keeping with classical geometry, many of the exercises that follow are phrased in terms of points rather than in terms of vectors.

In Exercises 1–6, sketch the indicated translate of the subset of $\mathbb{R}^n$ in an appropriate figure.

1. The translate of the line $x_2 = 2x_1 + 3$ in $\mathbb{R}^2$ by the vector $[-3, 0]$
2. The translate of $\{(t, t^2) \mid t \in \mathbb{R}\}$ in $\mathbb{R}^2$ by the vector $[-1, -2]$
3. The translate of $\{x \in \mathbb{R}^2 \mid |x_i| \leq 1 \text{ for } i = 1, 2\}$ by the vector $[1, 2]$
4. The translate of $\{x \in \mathbb{R}^2 \mid \|x\| \geq 3\}$ by the vector $[2, 3]$
5. The translate of $\{x \in \mathbb{R}^3 \mid \|x\| \leq 1\}$ by the vector $[2, 4, 3]$
6. The translate of the plane $x_1 + x_2 = 2$ in $\mathbb{R}^3$ by the vector $[-1, 2, 3]$
7. Give parametric equations for the line in $\mathbb{R}^2$ through $(3, -3)$ with direction vector $d = [-8, 4]$. Sketch the line in an appropriate figure.
8. Give parametric equations for the line in $\mathbb{R}^3$ through $(-1, 3, 0)$ with direction vector $d = [-2, -1, 4]$. Sketch the line in an appropriate figure.
9. Consider the line in $\mathbb{R}^2$ that is given by the equation $d_1x_1 + d_2x_2 = c$ for numbers d_1, d_2 , and c in $\mathbb{R}$, where d_1 and d_2 are not both zero. Find parametric equations of the line.
10. Find parametric equations for the line in $\mathbb{R}^2$ through $(5, -1)$ and orthogonal to the line with parametric equations $x_1 = 4 - 2t$, $x_2 = 7 + t$.
11. For each pair of points, find parametric equations of the line containing them.
 - a. $(-2, 4)$ and $(3, -1)$ in $\mathbb{R}^2$
 - b. $(3, -1, 6)$ and $(0, -3, -1)$ in $\mathbb{R}^3$
 - c. $(2, 0, 4)$ and $(-1, 5, -8)$ in $\mathbb{R}^3$
12. For each of the given pairs of lines in $\mathbb{R}^3$, determine whether the lines intersect. If they do intersect, find the point of intersection, and determine whether the lines are orthogonal.
 - a. $x_1 = 4 + t, \quad x_2 = 2 - 3t,$
 $x_3 = -3 + 5t$
 and
 $x_1 = 11 + 3s, \quad x_2 = -9 - 4s,$
 $x_3 = -4 - 3s$
 - b. $x_1 = 11 + 3t, \quad x_2 = -3 - t,$
 $x_3 = 4 + 3t$
 and
 $x_1 = 6 - 2s, \quad x_2 = -2 + s,$
 $x_3 = -15 + 7s$
13. Find all points in common to the lines in $\mathbb{R}^2$ given by $x_1 = 5 - 3t, x_2 = -1 + t$ and $x_1 = -7 + 6s, x_2 = 3 - 2s$.
14. Find parametric equations for the line in $\mathbb{R}^3$ through $(-1, 2, 3)$ that is orthogonal to each of the two lines having parametric equations $x_1 = -2 + 3t, x_2 = 4, x_3 = 1 - t$ and $x_1 = 7 - t, x_2 = 2 + 3t, x_3 = 4 + t$.
15. Find the midpoint of the line segment joining each pair of points.
 - a. $(-2, 4)$ and $(3, -1)$ in $\mathbb{R}^2$
 - b. $(3, -1, 6)$ and $(0, -3, -1)$ in $\mathbb{R}^3$
 - c. $(0, 4, 8)$ and $(-4, 5, 9)$ in $\mathbb{R}^3$
16. Find the point in $\mathbb{R}^2$ on the line segment joining $(-1, 3)$ and $(2, 5)$ that is twice as close to $(-1, 3)$ as to $(2, 5)$.

17. Find the point in $\mathbb{R}^3$ on the line segment joining $(-2, 1, 3)$ and $(0, -5, 6)$ that is one-fourth of the way from $(-2, 1, 3)$ to $(0, -5, 6)$.
 18. Find the points that divide the line segment between $(2, 1, 3, 4)$ and $(-1, 2, 1, 3)$ in $\mathbb{R}^4$ into three equal parts.
 19. Find the midpoint of the line segment between $(2, 1, 3, 4, 0)$ and $(1, 2, -1, 3, -1)$ in $\mathbb{R}^5$.
 20. Find the intersection in $\mathbb{R}^3$ of the line given by

$$x_1 = 5 + t, \quad x_2 = -3t, \quad x_3 = -2 + 4t$$
 and the plane with equation $x_1 - 3x_2 + 2x_3 = -25$.
 21. Find the intersection in $\mathbb{R}^3$ of the line given by

$$x_1 = 2, \quad x_2 = 5 - t, \quad x_3 = 2t$$
 and the plane with equation $x_1 + 2x_3 = 10$.
 22. Find parametric equations of the plane that passes through the unit coordinate points $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ in $\mathbb{R}^3$.
 23. Find a single linear equation in three variables whose solution set is the plane in Exercise 22. [HINT: We suggest two general methods of attack: (1) eliminate the parameters from your answer to Exercise 22, or (2) solve an appropriate linear system. Actually, this particular answer can be found by inspection.]
 24. Find parametric equations of the plane in $\mathbb{R}^3$ that passes through $(1, 0, 0)$, $(0, 1, -1)$, and $(1, 1, 1)$.
 25. Find a single linear equation in three variables whose solution set is the plane in Exercise 24. [See the hint for Exercise 23.]
 26. Find a vector equation of the plane that passes through the points $(1, 2, 1)$, $(-1, 2, 3)$, and $(2, 1, 4)$ in $\mathbb{R}^3$.
 27. Find a single linear equation in three variables whose solution set is the plane in Exercise 26. [See the hint for Exercise 23.]
 28. Find a vector equation for the plane in $\mathbb{R}^4$ that passes through the points $(1, 2, 1, 3)$, $(4, 1, 2, 1)$, and $(3, 1, 2, 0)$.
 29. Find a linear system with two equations in four variables whose solution set is the plane in Exercise 28. [See the hint for Exercise 23.]
 30. Find a vector equation of the hyperplane that passes through the points $(1, 2, 1, 2, 3)$, $(0, 1, 2, 1, 3)$, $(0, 0, 3, 1, 2)$, $(0, 0, 0, 1, 4)$, and $(0, 0, 0, 0, 2)$ in $\mathbb{R}^5$.
 31. Find a single linear equation in five variables whose solution set is the hyperplane in Exercise 30. [See the hint for Exercise 23.]
 32. Find a vector equation of the hyperplane in $\mathbb{R}^6$ through the endpoints of $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_6$.
 33. Find a single linear equation in six variables whose solution set is the hyperplane in Exercise 32. [See the hint for Exercise 23.]
- In Exercises 34–42, solve the given system of linear equations and write the solution set as a k -flat.*
34. $x_1 - 2x_2 = 3$
 $3x_1 - x_2 = 14$
 35. $x_1 + 2x_2 - x_3 = -3$
 $3x_1 + 7x_2 + 2x_3 = 1$
 $4x_1 - 2x_2 + x_3 = -2$
 36. $x_1 + 4x_2 - 2x_3 = 4$
 $2x_1 + 7x_2 - x_3 = -2$
 $x_1 + 3x_2 + x_3 = -6$
 37. $x_1 - 3x_2 + x_3 = 2$
 $3x_1 - 8x_2 + 2x_3 = 5$
 $3x_1 - 7x_2 + x_3 = 4$
 38. $x_1 - 3x_2 + 2x_3 - x_4 = 8$
 $3x_1 - 7x_2 + x_4 = 0$
 39. $x_1 - 2x_3 + x_4 = 6$
 $2x_1 - x_2 + x_3 - 3x_4 = 0$
 $9x_1 - 3x_2 - x_3 - 7x_4 = 4$
 40. $x_1 + 2x_2 - 3x_3 + x_4 = 2$
 $3x_1 + 6x_2 - 8x_3 - 2x_4 = 1$

41. $x_1 - 3x_2 + x_3 + 2x_4 = 2$

$x_1 - 2x_2 + 2x_3 + 4x_4 = -1$

$2x_1 - 8x_2 - x_3 = 3$

$3x_1 - 9x_2 + 4x_3 = 7$

42. $2x_1 - 5x_2 + x_3 - 10x_4 + 15x_5 = 60$

43. Mark each of the following True or False.

___ a. The solution set of a linear equation in x_1 and x_2 can be regarded as a hyperplane in $\mathbb{R}^2$.___ b. Every line and hyperplane in $\mathbb{R}^n$ intersect in a single point.___ c. The intersection of two distinct hyperplanes in $\mathbb{R}^5$ is a line, if the intersection is nonempty.___ d. The Euclidean space $\mathbb{R}^5$ has no physical existence, but exists only in our minds.___ e. The Euclidean spaces $\mathbb{R}$, $\mathbb{R}^2$, and $\mathbb{R}^3$ have no physical existence, but exist only in our minds.

___ f. The mathematical existence of Euclidean 5-space is as substantial as the mathematical existence of Euclidean 3-space.

___ g. Every plane in $\mathbb{R}^n$ is a two-dimensional subspace of $\mathbb{R}^n$.___ h. Every plane through the origin in $\mathbb{R}^n$ is a two-dimensional subspace of $\mathbb{R}^n$.___ i. Every k -flat in $\mathbb{R}^n$ contains the origin.___ j. Every k -flat in $\mathbb{R}^n$ is a translate of a k -dimensional subspace.

VECTOR SPACES

For the sake of efficiency, mathematicians often study objects just in terms of their mathematical structure, deemphasizing such things as particular symbols used, names of things, and applications. Any properties derived exclusively from mathematical structure will hold for all objects having that structure. Organizing mathematics in this way avoids repeating the same arguments in different contexts. Viewed from this perspective, linear algebra is the study of all objects that have a *vector-space* structure. The Euclidean spaces $\mathbb{R}^n$ that we treated in Chapters 1 and 2 serve as our guide.

Section 3.1 defines the general notion of a *vector space*, motivated by the familiar algebraic structure of the spaces $\mathbb{R}^n$. Our examples focus mainly on spaces other than $\mathbb{R}^n$, such as function spaces. Unlike the first two chapters in our text, this chapter draws on calculus for many of its illustrations.

Section 3.2 explains how the linear-algebra terminology introduced in Chapter 1 for $\mathbb{R}^n$ carries over to a general vector space V . The definitions given in Chapters 1 and 2 for linear combinations, spans, subspaces, bases, dependent vectors, independent vectors, and dimension can be left mostly unchanged, except for replacing " $\mathbb{R}^n$ " by "a vector space V ." Indeed, with this replacement, many of the theorems and proofs in Chapter 1 have word-for-word validity for general vector spaces.

Section 3.3 shows that every finite-dimensional (real) vector space can be *coordinatized* to become algebraically indistinguishable from one of the spaces $\mathbb{R}^n$. This coordinatization allows us to apply the matrix techniques developed in Chapters 1 and 2 to any finite-dimensional vector space for such things as determining whether vectors are independent or form a basis.

Linear transformations of one vector space into another are the topic of Section 3.4. We will see that some of the basic operations of calculus, such as differentiation, can be viewed as linear transformations.

To conclude the chapter, optional Section 3.5 describes how we try to access such geometric notions as length and angle even in infinite-dimensional vector spaces.

3.1

VECTOR SPACES

The Vector-Space Operations

In each Euclidean space $\mathbb{R}^n$, we know how to add two vectors and how to perform scalar multiplication of a vector by a real number (scalar). These are the two *vector-space operations*. The first requirement that a set V , whose elements we will call “vectors,” must satisfy in order to be a vector space is that it have two well-defined algebraic operations, each of which yields an element of V —namely:

- | | |
|--|-----------------------|
| Addition of two elements of V | Vector addition |
| Multiplication of an element of V by a scalar. | Scalar multiplication |

For example, we know how to add the two functions x^2 and $\sin x$, and we know how to multiply them by a real number. We require that, whenever addition or scalar multiplication is performed with elements in V , *the answers obtained lie again in V* . That is, we require that V be **closed under vector addition** and **closed under scalar multiplication**. This notion of *closure* under an operation is familiar to us from Chapter 1.

Definition of a Vector Space

The definition of a *vector space* which follows incorporates the ideas we have just discussed. It also requires that the vector addition and scalar multiplication satisfy the algebraic properties that hold in $\mathbb{R}^n$ —namely, those listed in Theorem 1.1.

DEFINITION 3.1 Vector Space

A (real) **vector space** is a set V of objects called **vectors**, together with a rule for adding any two vectors v and w to produce a vector $v + w$ in V and a rule for multiplying any vector v in V by any scalar r in $\mathbb{R}$ to produce a vector rv in V . Moreover, there must exist a vector 0 in V , and for each v in V there must exist a vector $-v$ in V such that properties A1 through A4 and S1 through S4 below are satisfied for all choices of vectors $u, v, w \in V$ and scalars $r, s \in \mathbb{R}$.

Properties of Vector Addition

- | | |
|--------------------------------|---------------------------------|
| A1 $(u + v) + w = u + (v + w)$ | An associative law |
| A2 $v + w = w + v$ | A commutative law |
| A3 $0 + v = v$ | 0 as additive identity |
| A4 $v + (-v) = 0$ | $-v$ as additive inverse of v |

Properties Involving Scalar Multiplication

- | | | |
|----|----------------------|-----------------------|
| S1 | $r(v + w) = rv + rw$ | A distributive law |
| S2 | $(r + s)v = rv + sv$ | A distributive law |
| S3 | $r(sv) = (rs)v$ | An associative law |
| S4 | $1v = v$ | Preservation of scale |

In a moment we will show that there is only one vector 0 in V satisfying condition A3; this vector is called the **zero vector**. Similarly, we will see that the vector $-v$ in condition A4 is uniquely determined by v ; it is called the **additive inverse** of v , and is usually read “minus v .” We write $v - w$ for $v + (-w)$.

The adjective “real” in parentheses in the first line of Definition 3.1 signifies that it is sometimes necessary to allow the scalars to be complex numbers, $a + bi$ where $a, b \in \mathbb{R}$ and $i^2 = -1$. Linear algebra using complex scalars is the topic of Chapter 9. There are several branches of mathematics in which one does not gain full insight without using complex numbers, and linear algebra is one of them. Unfortunately, pencil and paper computation with complex numbers is cumbersome, and so it is customary in a first course to work mostly with real scalars. Fortunately, many of the concepts of linear algebra can be adequately explained in terms of real vector spaces.

Because our definition of a vector space was modeled on the algebraic structure of the Euclidean spaces $\mathbb{R}^n$ discussed in Chapter 1, we see that $\mathbb{R}^n$ is a vector space for each positive integer n . We now proceed to illustrate this concept with other examples.

HISTORICAL NOTE ALTHOUGH THE OBJECTS WE CALL VECTOR SPACES were well known in the late nineteenth century, the first mathematician to give an abstract definition of a vector space was Giuseppe Peano (1858–1932) in his *Calcolo Geometrico* of 1888. Peano’s aim in the book, as the title indicates, was to develop a geometric calculus. Such a calculus “consists of a system of operations analogous to those of algebraic calculus but in which the objects with which the calculations are performed are, instead of numbers, geometrical objects.” Much of the book consists of calculations dealing with points, lines, planes, and volumes. But in the ninth chapter, Peano defines what he called a *linear system*. This was a set of objects that was provided with operations of addition and scalar multiplication. These operations were to satisfy axioms A1–A4 and S1–S4 presented in this section. Peano also defined the *dimension* of a linear system to be the maximum number of linearly independent objects in the system and noted that the set of polynomial functions in one variable forms a linear system of infinite dimension.

Curiously, Peano’s work had no immediate effect on the mathematical community. The definition was even forgotten. It only entered the mathematical mainstream through the book *Space-Time-Matter* (1918) by Hermann Weyl (1885–1955). Weyl wrote this book as an introduction to Einstein’s general theory of relativity. In Chapter 1 he discusses the nature of a Euclidean space and, as part of that discussion, formulates the same standard axioms as Peano did earlier. He also gives a philosophic reason for adopting such a definition:

Not only in geometry, but to a still more astonishing degree in physics, has it become more and more evident that as soon as we have succeeded in unraveling fully the natural laws which govern reality, we find them to be expressible by mathematical relations of surpassing simplicity and architectonic perfection. . . . Analytical geometry [the axiom system which he presented] . . . conveys an idea, even if inadequate, of this perfection of form.

EXAMPLE 1 Show that the set $M_{m,n}$ of all $m \times n$ matrices is a vector space, using as vector addition and scalar multiplication the usual addition of matrices and multiplication of a matrix by a scalar.

SOLUTION We have seen that addition of $m \times n$ matrices and multiplication of an $m \times n$ matrix by a scalar again yield an $m \times n$ matrix. Thus, $M_{m,n}$ is closed under vector addition and scalar multiplication. We take as zero vector in $M_{m,n}$ the usual zero matrix, all of whose entries are zero. For any matrix A in $M_{m,n}$ we consider $-A$ to be the matrix $(-1)A$. The properties of matrix arithmetic on page 45 show that all eight properties A1–A4 and S1–S4 required of a vector space are satisfied. ■

The preceding example introduced the notation $M_{m,n}$ for the vector space of all $m \times n$ matrices. We use M_n for the vector space of all square $n \times n$ matrices.

EXAMPLE 2 Show that the set P of all polynomials in the variable x with coefficients in $\mathbb{R}$ is a vector space, using for vector addition and scalar multiplication the usual addition of polynomials and multiplication of a polynomial by a scalar.

SOLUTION Let p and q be polynomials

$$p = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

and

$$q = b_0 + b_1x + b_2x^2 + \cdots + b_mx^m.$$

If $m \geq n$, the **sum** of p and q is given by

$$p + q = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n + b_{n+1}x^{n+1} + \cdots + b_mx^m.$$

For example, if $p = 1 + 2x + 3x^2$ and $q = x + x^3$, then $p + q = 1 + 3x + 3x^2 + x^3$. A similar definition is made if $m < n$. The **product** of p and a scalar r is given by

$$rp = ra_0 + ra_1x + ra_2x^2 + \cdots + ra_nx^n.$$

Taking the usual notions of the zero polynomial and of $-p$, we recognize that the eight properties A1–A4 and S1–S4 required of a vector space are familiar properties for these polynomial operations. Thus, P is a vector space. ■

EXAMPLE 3 Let F be the set of all real-valued functions with domain $\mathbb{R}$; that is, let F be the set of all functions mapping $\mathbb{R}$ into $\mathbb{R}$. The vector **sum** $f + g$ of two functions f and g in F is defined in the usual way to be the function whose value at any x in $\mathbb{R}$ is $f(x) + g(x)$; that is,

$$(f + g)(x) = f(x) + g(x).$$

For any scalar r in $\mathbb{R}$ and function f in F , the **product** rf is the function whose value at x is $rf(x)$, so that

$$(rf)(x) = rf(x).$$

Show that F with these operations is a vector space.

SOLUTION We observe that, for f and g in F , both $f + g$ and rf are functions mapping $\mathbb{R}$ into $\mathbb{R}$, so $f + g$ and rf are in F . Thus, F is closed under vector addition and under scalar multiplication. We take as zero vector in F the constant function whose value at each x in $\mathbb{R}$ is 0. For each function f in F , we take as $-f$ the function $(-1)f$ in F .

There are four vector-addition properties to verify, and they are all easy. We illustrate by verifying condition A4. For f in F , the function $f + (-f) = f + (-1)f$ has as its value at x in $\mathbb{R}$ the number $f(x) + (-1)f(x)$, which is 0. Consequently, $f + (-f)$ is the zero function, and A4 is verified.

The scalar multiplicative properties are just as easy to verify. For example, to verify S4, we must compute $1f$ at any x in $\mathbb{R}$ and compare the result with $f(x)$. We obtain $(1f)(x) = 1f(x) = f(x)$, and so $1f = f$. ■

EXAMPLE 4 Show that the set P_∞ of *formal power series* in x of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n + \cdots,$$

with addition and scalar multiplication defined by

$$\left(\sum_{n=0}^{\infty} a_n x^n \right) + \left(\sum_{n=0}^{\infty} b_n x^n \right) = \sum_{n=0}^{\infty} (a_n + b_n) x^n \quad \text{and} \quad r \left(\sum_{n=0}^{\infty} a_n x^n \right) = \sum_{n=0}^{\infty} r a_n x^n,$$

is a vector space.*

SOLUTION The reasoning here is precisely the same as for the space P of polynomials in Example 2. The zero series is $\sum_{n=0}^{\infty} 0x^n$, and the additive inverse of $\sum_{n=0}^{\infty} a_n x^n$ is $\sum_{n=0}^{\infty} (-a_n) x^n$. All the other axioms follow from the associative, commutative, and distributive laws for the real-number coefficients a_n in the series. ■

The examples of vector spaces that we have presented so far are all based on algebraic structures familiar to us—namely, the algebra of matrices, polynomials, and functions. You may be thinking, “How can anything with an

*We can add only a *finite* number of vectors in a vector space. Thus we do not regard these formal power series as infinite sums in P_∞ of monomials. Also, we are not concerned with questions of convergence or divergence, as studied in calculus. This is the significance of the word *formal*.

addition and scalar multiplication defined on it fail to be a vector space?" To answer this, we give two more-esoteric examples, where checking the axioms is not as natural a process.

EXAMPLE 5 Let $\mathbb{R}^2$ have the usual operation of addition, but define scalar multiplication $r[x, y]$ by $r[x, y] = [0, 0]$. Determine whether $\mathbb{R}^2$ with these operations is a vector space.

SOLUTION Because conditions A1–A4 of Definition 3.1 do not involve scalar multiplication, and because addition is the usual operation, we need only check conditions S1–S4. We see that all of these hold, except for condition S4: because $1[x, y] = [0, 0]$, the scale is not preserved. Thus, $\mathbb{R}^2$ is not a vector space with these particular two operations. ■

EXAMPLE 6 Let $\mathbb{R}^2$ have the usual scalar multiplication, but let addition $\dot{+}$ be defined on $\mathbb{R}^2$ by the formula

$$[x, y] \dot{+} [r, s] = [x + r, 2y + s].$$

Determine whether $\mathbb{R}^2$ with these operations is a vector space. (We use the symbol $\dot{+}$ for the warped addition of vectors in $\mathbb{R}^2$, to distinguish it from the usual addition.)

SOLUTION We check the associative law for $\dot{+}$:

$$\begin{aligned} ([x, y] \dot{+} [r, s]) \dot{+} [a, b] &= [x + r, 2y + s] \dot{+} [a, b] \\ &= [(x + r) + a, 2(2y + s) + b] \\ &= [x + r + a, 4y + 2s + b], \end{aligned}$$

whereas

$$\begin{aligned} [x, y] \dot{+} ([r, s] \dot{+} [a, b]) &= [x, y] \dot{+} [r + a, 2s + b] \\ &= [x + (r + a), 2y + (2s + b)] \\ &= [x + r + a, 2y + 2s + b]. \end{aligned}$$

Because the two colored scalars are not equal, we expect that $\dot{+}$ is not associative. We can find a specific violation of the associative law by choosing $y \neq 0$; for example,

$$([0, 1] \dot{+} [0, 0]) \dot{+} [0, 0] = [0, 4],$$

whereas

$$[0, 1] \dot{+} ([0, 0] \dot{+} [0, 0]) = [0, 2].$$

Therefore, $\mathbb{R}^2$ is not a vector space with these two operations. ■

We now indicate that vector addition and scalar multiplication possess still more of the properties we are accustomed to expect. It is important to realize that everything has to be proved using just the axioms A1 through A4 and S1 through S4. Of course, once we have proved something from the

axioms, we can then use it in the proofs of other things. The properties that appear in Theorem 3.1 below are listed in a convenient order for proof; for example, we will see that it is convenient to know property 3 in order to prove property 4. Property 4 states that in a vector space V , we have $0v = \mathbf{0}$ for all $v \in V$. Students often attempt to prove this by saying,

$$\begin{aligned} \text{"Let } v &= (a_1, a_2, \dots, a_n). \text{ Then } 0v = 0(a_1, a_2, \dots, a_n) \\ &= (0, 0, \dots, 0) = \mathbf{0}." \end{aligned}$$

This is a fine argument if V is $\mathbb{R}^n$, but we have now expanded our concept of *vector space*, and we can no longer assume that $v \in V$ is some n -tuple of real numbers.

THEOREM 3.1 Elementary Properties of Vector Spaces

Every vector space V has the following properties:

1. The vector $\mathbf{0}$ is the *unique* vector x satisfying the equation $x + v = v$ for all vectors v in V .
2. For each vector v in V , the vector $-v$ is the *unique* vector y satisfying $v + y = \mathbf{0}$.
3. If $u + v = u + w$ for vectors u, v , and w in V , then $v = w$.
4. $0v = \mathbf{0}$ for all vectors v in V .
5. $r\mathbf{0} = \mathbf{0}$ for all scalars r in $\mathbb{R}$.
6. $(-r)v = r(-v) = -(rv)$ for all scalars r in $\mathbb{R}$ and vectors v in V .

PROOF We prove only properties 1 and 4, leaving proofs of the remaining properties as Exercises 19 through 22. In proving property 4, we assume that properties 2 and 3 have been proved.

Turning to property 1, the standard way to prove that something is unique is to suppose that there are two of them, and then show that they must be equal. Suppose, therefore, that there exist vectors $\mathbf{0}$ and $\mathbf{0}'$ satisfying

$$\mathbf{0} + v = v \quad \text{and} \quad \mathbf{0}' + v = v \quad \text{for all } v \in V.$$

Taking $v = \mathbf{0}'$ in the first equation and $v = \mathbf{0}$ in the second equation, we obtain

$$\mathbf{0} + \mathbf{0}' = \mathbf{0}' \quad \text{and} \quad \mathbf{0}' + \mathbf{0} = \mathbf{0}.$$

By the commutative law A2, we know that $\mathbf{0} + \mathbf{0}' = \mathbf{0}' + \mathbf{0}$, and we conclude that $\mathbf{0} = \mathbf{0}'$.

Turning to property 4, notice that the equation $0v = \mathbf{0}$ which we want to prove involves both scalar multiplication (namely, $0v$) and vector addition ($\mathbf{0}$ is an *additive* concept, given by axiom A3). To prove a relationship between these two algebraic operations, we must use an axiom that involves *both* of

them—namely, one of the distributive laws S1 or S2. Using distributive law S2, we have for $v \in V$,

$$0v = (0 + 0)v = 0v + 0v.$$

By the additive identity axiom A3 and the commutative law A2, we know that

$$0v = 0 + 0v = 0v + 0.$$

Therefore,

$$0v + 0v = 0v + 0,$$

and, by property (3), we conclude that $0v = 0$. $\blacktriangle$

The Universality of Function Spaces (Optional)

In Example 3, we showed that the set F of all functions mapping $\mathbb{R}$ into $\mathbb{R}$ is a vector space, where we define for $f, g \in F$ and for $r \in \mathbb{R}$

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (rf)(x) = rf(x). \quad (1)$$

Note that these definitions of addition and scalar multiplication for functions having $\mathbb{R}$ as both domain and codomain use only the algebraic structure of the codomain $\mathbb{R}$ and not the algebraic structure of the domain $\mathbb{R}$. That is, the defining addition and scalar multiplication appearing on the right-hand sides of Eqs. (1) take place in the codomain. We do not see anything like $f(a + b)$, which involves addition in the domain $\mathbb{R}$, or like $f(rx)$, which involves scalar multiplication in the domain. This suggests that if S is any set and we let F be the set of all functions mapping S into $\mathbb{R}$, then Example 3 might still go through, and show that F is a vector space. We show that this is the case in our next example.

EXAMPLE 7 Let F be the set of all real-valued functions on a (nonempty) set S ; that is, let F be the set of all functions mapping S into $\mathbb{R}$. For $f, g \in F$, let the sum $f + g$ of two functions f and g in F be defined by

$$(f + g)(x) = f(x) + g(x) \quad \text{for all } x \in S,$$

and, for any scalar r , let scalar multiplication be defined by

$$(rf)(x) = rf(x) \quad \text{for all } x \in S.$$

Show that F with these operations is a vector space.

SOLUTION The solution of Example 3 is valid word for word if we replace each reference $\mathbb{R}$ to the domain by the new domain S . That is, the functions map S into $\mathbb{R}$, and we let $x \in S$ rather than $x \in \mathbb{R}$ in this solution. $\blacksquare$

We now indicate why we headed this discussion “The Universality of Function Spaces.” Let $S = \{1, 2\}$, so that F is the set of all functions mapping $\{1, 2\}$ into $\mathbb{R}$. Let us abbreviate the description of a function f as $[f(1), f(2)]$. For

example, we consider $[-3, 8]$ to denote the function in F that maps 1 into -3 and maps 2 into 8. In this way, we identify each vector $[a, b]$ in $\mathbb{R}^2$ with a function f mapping $\{1, 2\}$ into $\mathbb{R}$ —namely, $f(1) = a$ and $f(2) = b$. If we view $[a, b]$ as the function f and $[c, d]$ as the function g , then

$$(f + g)(1) = f(1) + g(1) = a + c \quad \text{and} \quad (f + g)(2) = f(2) + g(2) = b + d.$$

Thus, the vector $[a + c, b + d] = [a, b] + [c, d]$ in $\mathbb{R}^2$ is the function $f + g$. Similarly, we see that the vector $[ra, rb] = r[a, b]$ in $\mathbb{R}^2$ is the function rf . In this way, we can regard $\mathbb{R}^2$ as the vector space of functions mapping $\{1, 2\}$ into $\mathbb{R}$. We realize that we can equally well consider each vector of $\mathbb{R}^n$ as a function mapping $\{1, 2, 3, \dots, n\}$ into $\mathbb{R}$. For example, the vector $[-3, 5, 2, 7]$ can be considered as the function $f: \{1, 2, 3, 4\} \rightarrow \mathbb{R}$, where $f(1) = -3, f(2) = 5, f(3) = 2$, and $f(4) = 7$.

In a similar fashion, we can view the vector space $M_{m,n}$ of matrices in Example 1 as the vector space of functions mapping the positive integers from 1 to mn into $\mathbb{R}$. For example, taking $m = 2$ and $n = 3$, we can view the matrix

$$\begin{bmatrix} a_1 & a_3 & a_5 \\ a_2 & a_4 & a_6 \end{bmatrix}$$

as the function $f: \{1, 2, 3, 4, 5, 6\} \rightarrow \mathbb{R}$, where $f(i) = a_i$. Addition of functions as defined in Example 7 again corresponds to addition of matrices, and the same is true for scalar multiplication.

The vector space P of all polynomials in Example 2 is not quite as easy to present as a function space, because not all the polynomials have the same number of terms. However, we can view the vector space P_∞ of formal power series in x as the space of all functions mapping $\{0, 1, 2, 3, \dots\}$ into $\mathbb{R}$. Namely, if f is such a function and if $f(n) = a_n$ for $n \in \{0, 1, 2, 3, \dots\}$, then we

can denote this function symbolically by $\sum_{n=0}^{\infty} a_n x^n$. We see that function

addition and multiplication by a scalar will produce precisely the addition and multiplication of power series defined in Example 4. We will show in the next section that we can view the vector space P of polynomials as a *subspace* of P_∞ .

We have now freed the domain of our function spaces from having to be the set $\mathbb{R}$. Let's see how much we can free the codomain. The definitions

$$(f + g)(x) = f(x) + g(x) \quad \text{and} \quad (rf)(x) = rf(x)$$

for addition and scalar multiplication of functions show that we need a notion of addition and scalar multiplication for the codomain. For function addition to be associative and commutative, we need the addition in the codomain to be associative and commutative. For there to be a zero vector, we need an additive identity—let's call it 0 —in the codomain, so that we will have a “zero constant function” to serve as additive identity, etc. It looks as though what we need is to have the codomain itself have a vector space structure! This is indeed the case. If S is a set and V is a vector space, then the set F of all functions $f: S \rightarrow V$ with this notion of function addition and multiplication by

a scalar is again a vector space. Note that $\mathbb{R}$ itself is a vector space, so the set of functions $f: S \rightarrow \mathbb{R}$ in Example 7 is a special case of this construction. For another example, the set of all functions mapping $\mathbb{R}^3$ into $\mathbb{R}^3$ has a vector space structure. In third-semester calculus, we sometimes speak of a “vector-valued function,” meaning a function whose codomain is a vector space, although we usually don’t talk about vector spaces there.

Thus, starting with a set S , we could consider the vector space V_1 of all functions mapping S into $\mathbb{R}$, and then the vector space V_2 of all functions mapping S into V_1 , and then the vector space V_3 of all functions mapping S into V_2 , and then—oops! We had better stop now. People who do too much of this stuff are apt to start climbing the walls. (However, mathematicians do sometimes build cumulative structures in this fashion.)

SUMMARY

1. A vector space is a nonempty set V of objects called *vectors*, together with rules for adding any two vectors $\mathbf{v}$ and $\mathbf{w}$ in V and for multiplying any vector $\mathbf{v}$ in V by any scalar r in $\mathbb{R}$. Furthermore, V must be closed under this vector addition and scalar multiplication so that $\mathbf{v} + \mathbf{w}$ and $r\mathbf{v}$ are both in V . Moreover, the following axioms must be satisfied for all vectors $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ in V and all scalars r and s in $\mathbb{R}$:

$$A1 \quad (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$A2 \quad \mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$$

$$A3 \quad \text{There exists a zero vector } \mathbf{0} \text{ in } V \text{ such that } \mathbf{0} + \mathbf{v} = \mathbf{v} \text{ for all } \mathbf{v} \in V.$$

$$A4 \quad \text{Each } \mathbf{v} \in V \text{ has an additive inverse } -\mathbf{v} \text{ in } V \text{ such that } \mathbf{v} + (-\mathbf{v}) = \mathbf{0}.$$

$$S1 \quad r(\mathbf{v} + \mathbf{w}) = r\mathbf{v} + r\mathbf{w}$$

$$S2 \quad (r + s)\mathbf{v} = r\mathbf{v} + s\mathbf{v}$$

$$S3 \quad r(s\mathbf{v}) = (rs)\mathbf{v}$$

$$S4 \quad 1\mathbf{v} = \mathbf{v}$$

2. Elementary properties of vector spaces are listed in Theorem 3.1.
3. Examples of vector spaces include $\mathbb{R}^n$, the space $M_{m,n}$ of all $m \times n$ matrices, the space of all polynomials in the variable x , and the space of all functions $f: \mathbb{R} \rightarrow \mathbb{R}$. For all of these examples, vector addition and scalar multiplication are the addition and multiplication by a real number with which we are already familiar.
4. (Optional) For any set S , the set of all functions mapping S into $\mathbb{R}$ forms a vector space with the usual definitions of addition and scalar multiplication of functions. The Euclidean space $\mathbb{R}^n$ can be viewed as the vector space of functions mapping $\{1, 2, 3, \dots, n\}$ into $\mathbb{R}$, where a vector $\mathbf{a} = [a_1, a_2, a_3, \dots, a_n]$ is viewed as the function f such that $f(i) = a_i$.
5. (Optional) Still more generally, for any set S and any vector space V , the set of all functions mapping S into V forms a vector space with the usual definitions of addition and scalar multiplication of functions.